

Representation Learning With PCA and Autoencoders

Group Number: 123

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This notebook solves the assignment tasks for MNIST and CIFAR10 converted to gray level.

Requirements implemented:

- MNIST and CIFAR10 loaded from `tensorflow.keras.datasets`
- CIFAR10 converted to gray level
- both datasets resized to 28 x 28
- intensities rescaled to [50, 200]
- random 70% / 20% / 10% train / validation / test split
- standard PCA and randomized PCA with 30 components
- logistic regression classification on PCA features and autoencoder features
- multiclass ROC curves
- reconstruction SNR in dB
- tied-weight linear autoencoder with unit-magnitude encoder vectors
- component/weight visualization and quantitative PCA-AE comparison
- deep convolutional autoencoder with 30-dimensional latent space
- single hidden layer and three hidden layer dense autoencoder comparisons

Before running:

```
# Uncomment if your environment is missing packages.
# %pip install -q tensorflow scikit-learn matplotlib pandas seaborn

import os
import socket
import random
from datetime import datetime
import warnings
```

```

from dataclasses import dataclass

# Keep Matplotlib cache inside a writable location when running from
# managed kernels.
os.environ.setdefault("MPLCONFIGDIR", os.path.join("/tmp",
"matplotlib"))
os.makedirs(os.environ["MPLCONFIGDIR"], exist_ok=True)

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display

import sklearn
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    roc_auc_score,
    roc_curve,
    auc,
)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import label_binarize, StandardScaler

from scipy.optimize import linear_sum_assignment

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

warnings.filterwarnings("ignore")
sns.set_theme(style="whitegrid")

```

Configuration

`FAST_MODE=False` runs the full 70/20/10 split required by the assignment. If you only want to verify that the notebook executes, set `FAST_MODE=True`.

```

RANDOM_SEED = 42
FAST_MODE = False

N_COMPONENTS = 30
IMAGE_SHAPE = (28, 28)
N_CLASSES = 10

BATCH_SIZE = 256

```

```

LINEAR_AE_EPOCHS = 150
DENSE_AE_EPOCHS = 60
CONV_AE_EPOCHS = 50
PATIENCE = 12

if FAST_MODE:
    # Small values for quick testing only. Leave FAST_MODE=False for
    # submission runs.
    MAX_TOTAL_PER_DATASET = 12000
    LINEAR_AE_EPOCHS = 10
    DENSE_AE_EPOCHS = 8
    CONV_AE_EPOCHS = 8
else:
    MAX_TOTAL_PER_DATASET = None

np.random.seed(RANDOM_SEED)
random.seed(RANDOM_SEED)
tf.random.set_seed(RANDOM_SEED)

print("TensorFlow:", tf.__version__)
print("scikit-learn:", sklearn.__version__)
print("FAST_MODE:", FAST_MODE)
gpus = tf.config.list_physical_devices("GPU")
print("GPU devices:", gpus)
for gpu in gpus:
    details = tf.config.experimental.get_device_details(gpu)
    print("GPU name:", details.get("device_name", "unknown"))
print("Hostname:", socket.gethostname())
print("Date/Time:", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))

TensorFlow: 2.20.0
scikit-learn: 1.6.1
FAST_MODE: False
GPU devices: [PhysicalDevice(name='/physical_device:GPU:0',
device_type='GPU')]
GPU name: Tesla T4
Hostname: 9f350ba3d94f
Date/Time: 2026-06-08 12:18:08

```

Helper Functions

```

@dataclass
class DatasetSplit:
    name: str
    x_train_img_50_200: np.ndarray
    y_train: np.ndarray
    x_val_img_50_200: np.ndarray
    y_val: np.ndarray
    x_test_img_50_200: np.ndarray
    y_test: np.ndarray

```

```

@property
def x_train_flat_50_200(self):
    return
self.x_train_img_50_200.reshape(len(self.x_train_img_50_200), -1)

@property
def x_val_flat_50_200(self):
    return
self.x_val_img_50_200.reshape(len(self.x_val_img_50_200), -1)

@property
def x_test_flat_50_200(self):
    return
self.x_test_img_50_200.reshape(len(self.x_test_img_50_200), -1)

@property
def x_train_01(self):
    return (self.x_train_img_50_200 - 50.0) / 150.0

@property
def x_val_01(self):
    return (self.x_val_img_50_200 - 50.0) / 150.0

@property
def x_test_01(self):
    return (self.x_test_img_50_200 - 50.0) / 150.0

@property
def x_train_flat_01(self):
    return self.x_train_01.reshape(len(self.x_train_01), -1)

@property
def x_val_flat_01(self):
    return self.x_val_01.reshape(len(self.x_val_01), -1)

@property
def x_test_flat_01(self):
    return self.x_test_01.reshape(len(self.x_test_01), -1)

def set_all_seeds(seed=RANDOM_SEED):
    np.random.seed(seed)
    random.seed(seed)
    tf.random.set_seed(seed)

def to_50_200(x):
    """Map image intensities from [0, 255] to [50, 200]."""
    x = x.astype("float32")

```

```

    return 50.0 + (x / 255.0) * 150.0

def preprocess_mnist_images(x):
    x = x.astype("float32")
    if x.ndim == 3:
        x = x[..., np.newaxis]
        # MNIST is already 28 x 28; keep this call for a consistent pipeline.
    x = tf.image.resize(x, IMAGE_SHAPE, method="bilinear").numpy()
    return to_50_200(x)

def preprocess_cifar10_images(x):
    x = x.astype("float32")
    # ITU-R BT.601 luminance conversion.
    gray = 0.299 * x[..., 0] + 0.587 * x[..., 1] + 0.114 * x[..., 2]
    gray = gray[..., np.newaxis]
    gray = tf.image.resize(gray, IMAGE_SHAPE,
method="bilinear").numpy()
    return to_50_200(gray)

def make_split(name, x, y, max_total=None):
    y = y.reshape(-1).astype("int64")

    if max_total is not None and max_total < len(x):
        rng = np.random.default_rng(RANDOM_SEED)
        idx = rng.choice(len(x), size=max_total, replace=False)
        x = x[idx]
        y = y[idx]

    # First split off 10% test, then split the remaining 90% into 70/20.
    x_temp, x_test, y_temp, y_test = train_test_split(
        x,
        y,
        test_size=0.10,
        random_state=RANDOM_SEED,
        stratify=y,
    )
    val_fraction_of_temp = 0.20 / 0.90
    x_train, x_val, y_train, y_val = train_test_split(
        x_temp,
        y_temp,
        test_size=val_fraction_of_temp,
        random_state=RANDOM_SEED,
        stratify=y_temp,
    )

```

```

    return DatasetSplit(name, x_train, y_train, x_val, y_val, x_test,
y_test)

def mean_snr_db(original, reconstructed, eps=1e-10):
    """Average per-image SNR in dB: 10 log10(signal_power /
noise_power)."""
    original = original.reshape(len(original), -1).astype("float64")
    reconstructed = reconstructed.reshape(len(reconstructed), -
1).astype("float64")
    signal_power = np.mean(original**2, axis=1)
    noise_power = np.mean((original - reconstructed) ** 2, axis=1)
    return float(np.mean(10.0 * np.log10((signal_power + eps) /
(noise_power + eps))))

def plot_component_grid(components, title, n_rows=5, n_cols=6,
image_shape=IMAGE_SHAPE, cmap="gray"):
    fig, axes = plt.subplots(n_rows, n_cols, figsize=(10, 8))
    fig.suptitle(title, fontsize=15, y=0.98)
    for i, ax in enumerate(axes.ravel()):
        comp = components[i].reshape(image_shape)
        ax.imshow(comp, cmap=cmap)
        ax.set_title(f"{i + 1}", fontsize=8)
        ax.axis("off")
    plt.tight_layout()
    plt.show()

def plot_reconstruction_examples(original, reconstructed, title, n=8):
    original = original[:n].reshape(n, *IMAGE_SHAPE)
    reconstructed = reconstructed[:n].reshape(n, *IMAGE_SHAPE)
    fig, axes = plt.subplots(2, n, figsize=(1.7 * n, 3.6))
    fig.suptitle(title, fontsize=14)
    for i in range(n):
        axes[0, i].imshow(original[i], cmap="gray", vmin=50, vmax=200)
        axes[0, i].axis("off")
        axes[0, i].set_title("Test", fontsize=8)
        axes[1, i].imshow(reconstructed[i], cmap="gray", vmin=50,
vmax=200)
        axes[1, i].axis("off")
        axes[1, i].set_title("Recon", fontsize=8)
    plt.tight_layout()
    plt.show()

def fit_logistic_and_roc(z_train, y_train, z_test, y_test, title):
    clf = LogisticRegression(
        max_iter=1500,
        solver="lbfgs",

```

```

        n_jobs=-1,
        random_state=RANDOM_SEED,
    )
    clf.fit(z_train, y_train)
    y_pred = clf.predict(z_test)
    y_score = clf.predict_proba(z_test)

    acc = accuracy_score(y_test, y_pred)
    macro_auc = roc_auc_score(y_test, y_score, multi_class="ovr",
average="macro")

    y_test_bin = label_binarize(y_test, classes=np.arange(N_CLASSES))
    fpr = {}
    tpr = {}
    roc_auc = {}
    for class_id in range(N_CLASSES):
        fpr[class_id], tpr[class_id], _ = roc_curve(y_test_bin[:,
class_id], y_score[:, class_id])
        roc_auc[class_id] = auc(fpr[class_id], tpr[class_id])

    plt.figure(figsize=(8, 6))
    for class_id in range(N_CLASSES):
        plt.plot(fpr[class_id], tpr[class_id], lw=1.2, label=f"Class
{class_id} AUC={roc_auc[class_id]:.3f}")
    plt.plot([0, 1], [0, 1], "k--", lw=1)
    plt.xlabel("False positive rate")
    plt.ylabel("True positive rate")
    plt.title(f"{title}\nAccuracy={acc:.4f}, macro
AUC={macro_auc:.4f}")
    plt.legend(fontsize=8, ncol=2)
    plt.tight_layout()
    plt.show()

    return clf, {"accuracy": acc, "macro_auc": macro_auc}

def compare_component_spaces(pca_components, ae_components,
dataset_name):
    """Compare PCA components and AE weight vectors.

    Both inputs should have shape (n_components, n_pixels).
    Sign is ignored because eigenvectors are sign-ambiguous.
    """
    pca_norm = pca_components / np.linalg.norm(pca_components, axis=1,
keepdims=True)
    ae_norm = ae_components / np.linalg.norm(ae_components, axis=1,
keepdims=True)

    abs_cos = np.abs(pca_norm @ ae_norm.T)
    row_ind, col_ind = linear_sum_assignment(-abs_cos)

```

```

    matched = abs_cos[row_ind, col_ind]

    singular_values = np.linalg.svd(pca_norm @ ae_norm.T,
compute_uv=False)
    principal_angles_deg =
np.degrees(np.arccos(np.clip(singular_values, -1, 1)))

    summary = {
        "dataset": dataset_name,
        "mean_matched_abs_cosine": float(np.mean(matched)),
        "median_matched_abs_cosine": float(np.median(matched)),
        "min_matched_abs_cosine": float(np.min(matched)),
        "mean_subspace_singular_value":
float(np.mean(singular_values)),
        "max_principal_angle_deg":
float(np.max(principal_angles_deg)),
    }
    return summary, abs_cos, matched, principal_angles_deg

def make_reconstruction_dataset(x, training=False):
    ds = tf.data.Dataset.from_tensor_slices(x.astype("float32"))
    if training:
        ds = ds.shuffle(min(len(x), 10000), seed=RANDOM_SEED,
reshuffle_each_iteration=True)
    return ds.batch(BATCH_SIZE).map(lambda batch: (batch, batch),
num_parallel_calls=tf.data.AUTOTUNE).prefetch(tf.data.AUTOTUNE)

def make_input_dataset(x):
    return
tf.data.Dataset.from_tensor_slices(x.astype("float32")).batch(BATCH_SI
ZE).prefetch(tf.data.AUTOTUNE)

def early_stopping():
    return keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=PATIENCE,
        restore_best_weights=True,
    )

def to_keras_images_01(x_01):
    return x_01.astype("float32")

def flat01_to_50_200(x_flat_01):
    x_flat_01 = np.clip(x_flat_01, 0.0, 1.0)

```



```
return 50.0 + 150.0 * x_flat_01
```

Load and Preprocess Datasets

```
print("Loading MNIST and CIFAR10 from Keras...")
(mnist_x_train_raw, mnist_y_train_raw), (mnist_x_test_raw,
mnist_y_test_raw) = keras.datasets.mnist.load_data()
(cifar_x_train_raw, cifar_y_train_raw), (cifar_x_test_raw,
cifar_y_test_raw) = keras.datasets.cifar10.load_data()

mnist_x_raw = np.concatenate([mnist_x_train_raw, mnist_x_test_raw],
axis=0)
mnist_y_raw = np.concatenate([mnist_y_train_raw, mnist_y_test_raw],
axis=0)
cifar_x_raw = np.concatenate([cifar_x_train_raw, cifar_x_test_raw],
axis=0)
cifar_y_raw = np.concatenate([cifar_y_train_raw, cifar_y_test_raw],
axis=0)

mnist_x = preprocess_mnist_images(mnist_x_raw)
cifar_x = preprocess_cifar10_images(cifar_x_raw)

mnist = make_split("MNIST", mnist_x, mnist_y_raw,
max_total=MAX_TOTAL_PER_DATASET)
cifar10_gray = make_split("CIFAR10-gray", cifar_x, cifar_y_raw,
max_total=MAX_TOTAL_PER_DATASET)
datasets = [mnist, cifar10_gray]

for ds in datasets:
    print(
        f"{ds.name}: train={len(ds.y_train)},
validation={len(ds.y_val)}, test={len(ds.y_test)}, "
        f"range=({ds.x_train_img_50_200.min():.1f},
{ds.x_train_img_50_200.max():.1f})"
    )
```

Loading MNIST and CIFAR10 from Keras...

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 ————— 2s 0us/step

Downloading data from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>

170498071/170498071 ————— 30s 0us/step

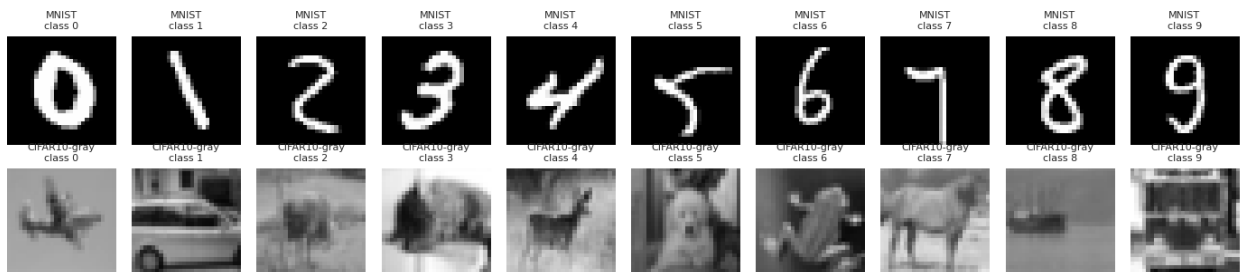
MNIST: train=48999, validation=14001, test=7000, range=(50.0, 200.0)

CIFAR10-gray: train=42000, validation=12000, test=6000, range=(50.0, 200.0)

```

fig, axes = plt.subplots(2, 10, figsize=(14, 3.2))
for row, ds in enumerate(datasets):
    for col in range(10):
        idx = np.where(ds.y_train == col)[0][0]
        axes[row, col].imshow(ds.x_train_img_50_200[idx].squeeze(),
cmap="gray", vmin=50, vmax=200)
        axes[row, col].set_title(f"{ds.name}\n\nclass {col}",
        fontsize=8)
        axes[row, col].axis("off")
plt.tight_layout()
plt.show()

```



Task 1: Standard PCA, Randomized PCA, Logistic Regression, ROC and SNR

For each dataset, PCA is fitted using the 70% training split. The first 30 principal components are then used as features for a multinomial logistic regression classifier. Reconstruction SNR is computed on the 10% test split.

The top-30 eigenvalues are tabulated explicitly after PCA fitting, and the matching component vectors are displayed as 28 x 28 grayscale images.

```

task1_rows = []
task1_models = {}

for ds in datasets:
    for solver_name, svd_solver in [("standard PCA", "full"),
    ("randomized PCA", "randomized")]:
        print(f"\n{ds.name}: fitting {solver_name}")
        pca = PCA(
            n_components=N_COMPONENTS,
            svd_solver=svd_solver,
            random_state=RANDOM_SEED,
            iterated_power=5 if svd_solver == "randomized" else
"auto",
        )

```

```

pca.fit(ds.x_train_flat_50_200)

z_train = pca.transform(ds.x_train_flat_50_200)
z_test = pca.transform(ds.x_test_flat_50_200)
recon_test = pca.inverse_transform(z_test)
snr = mean_snr_db(ds.x_test_flat_50_200, recon_test)

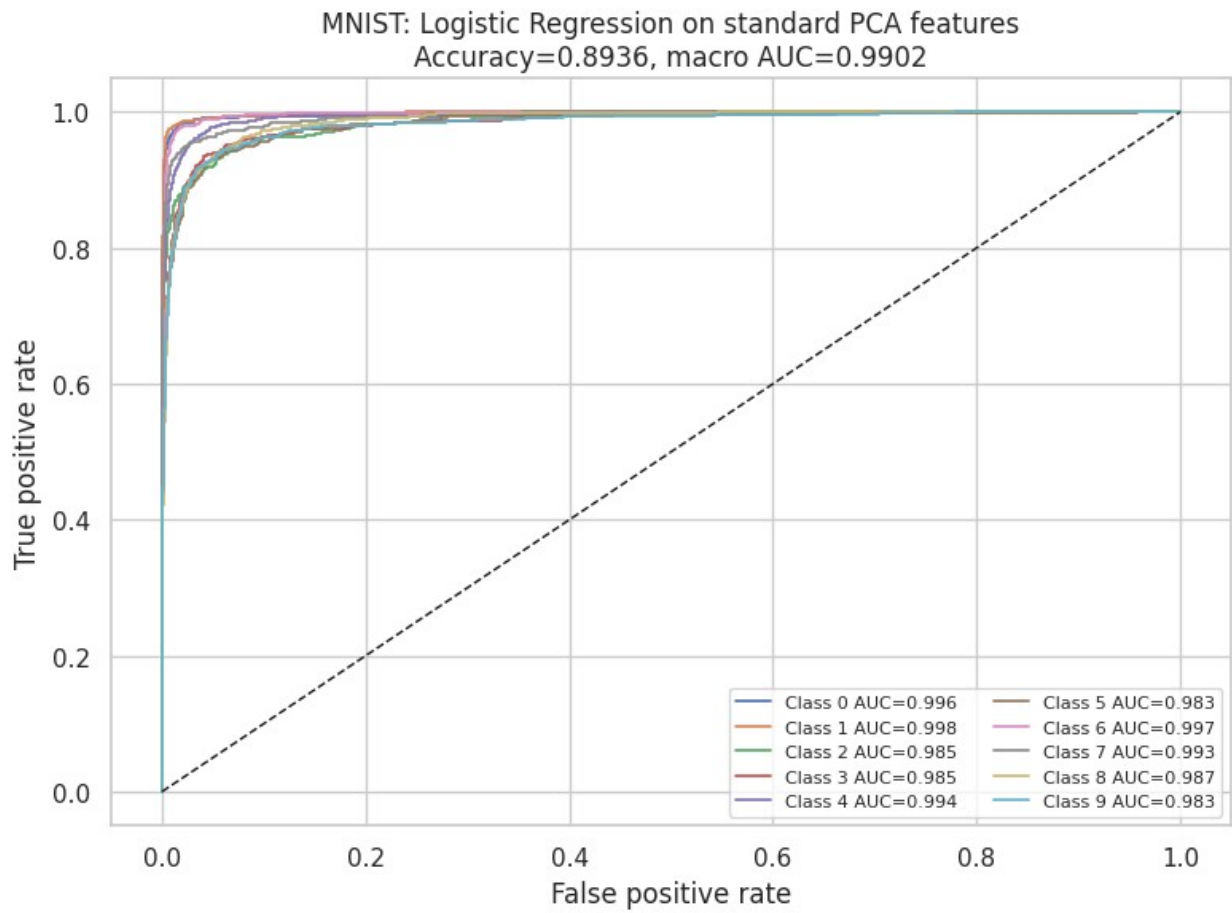
clf, metrics = fit_logistic_and_roc(
    z_train,
    ds.y_train,
    z_test,
    ds.y_test,
    title=f"{ds.name}: Logistic Regression on {solver_name}
features",
)

task1_rows.append(
    {
        "dataset": ds.name,
        "method": solver_name,
        "components": N_COMPONENTS,
        "accuracy": metrics["accuracy"],
        "macro_auc_ovr": metrics["macro_auc"],
        "explained_variance_ratio_sum":
float(np.sum(pca.explained_variance_ratio_)),
        "average_test_snr_db": snr,
    }
)
task1_models[(ds.name, solver_name)] = {
    "pca": pca,
    "clf": clf,
    "z_train": z_train,
    "z_test": z_test,
    "recon_test": recon_test,
}

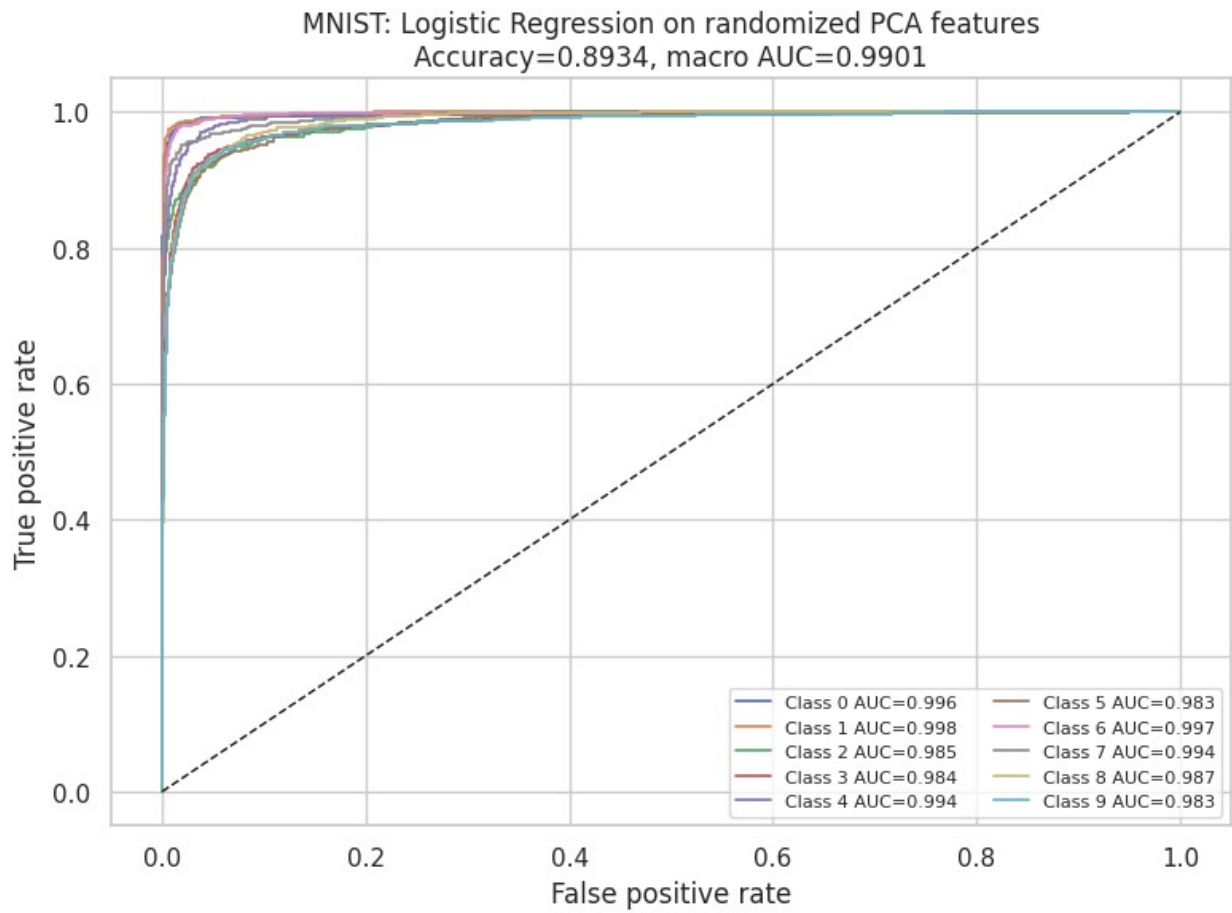
task1_results = pd.DataFrame(task1_rows)
task1_results

```

MNIST: fitting standard PCA

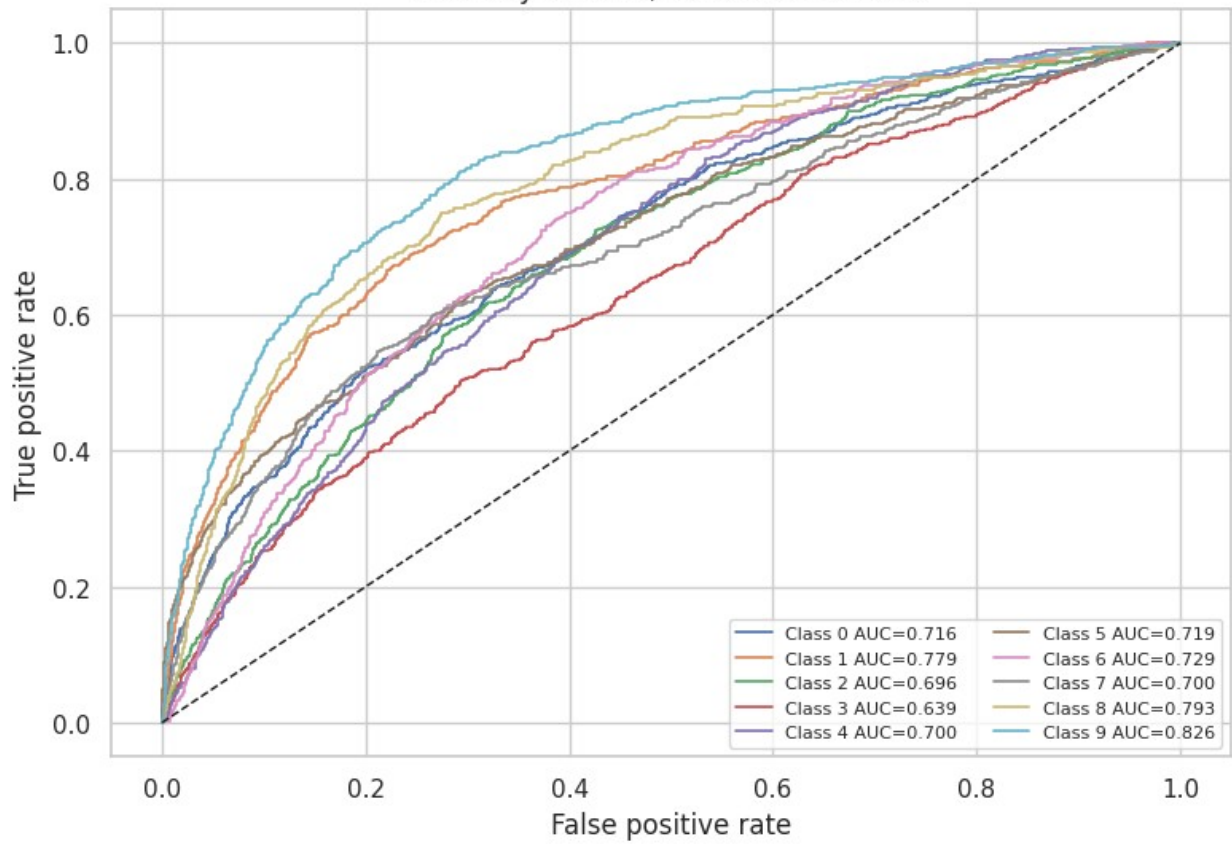


MNIST: fitting randomized PCA



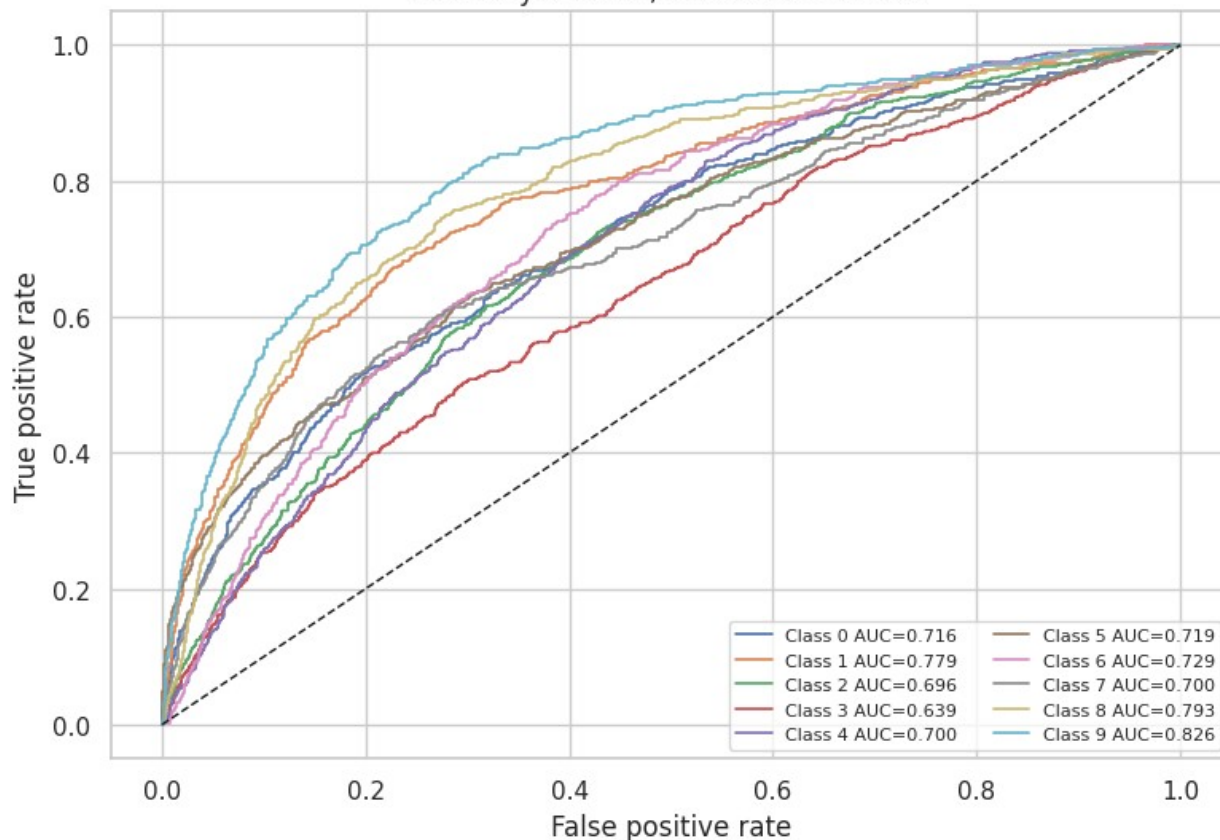
CIFAR10-gray: fitting standard PCA

CIFAR10-gray: Logistic Regression on standard PCA features
Accuracy=0.3070, macro AUC=0.7297



CIFAR10-gray: fitting randomized PCA

CIFAR10-gray: Logistic Regression on randomized PCA features
Accuracy=0.3072, macro AUC=0.7297



```
{
  "summary": {
    "name": "task1_results",
    "rows": 4,
    "fields": [
      {
        "column": "dataset",
        "dtype": "string",
        "num_unique_values": 2,
        "samples": [
          "CIFAR10-gray",
          "MNIST"
        ],
        "semantic_type": "",
        "description": ""
      },
      {
        "column": "method",
        "dtype": "string",
        "num_unique_values": 2,
        "samples": [
          "randomized PCA",
          "standard PCA"
        ],
        "semantic_type": "",
        "description": ""
      },
      {
        "column": "components",
        "dtype": "number",
        "std": 0,
        "min": 30,
        "max": 30,
        "num_unique_values": 1,
        "samples": [
          30
        ],
        "semantic_type": "",
        "description": ""
      },
      {
        "column": "accuracy",
        "dtype": "number",
        "std": 0.33856783221752135,
        "min": 0.307,
        "max": 0.8935714285714286,
        "num_unique_values": 4,
        "samples": [
          0.8934285714285715
        ],
        "semantic_type": ""
      }
    ]
  }
}
```

```

\ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ":
\ "macro_auc_ovr\ ",\n      \ "properties\ ": {\n      \ "dtype\ ":
\ "number\ ",\n      \ "std\ ": 0.15039327267694105,\n      \ "min\ ":
0.7296972530864198,\n      \ "max\ ": 0.9902317998587511,\n
\ "num_unique_values\ ": 4,\n      \ "samples\ ": [\n
0.9901408556435897\n      ],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ":
\ "explained_variance_ratio_sum\ ",\n      \ "properties\ ": {\n
\ "dtype\ ": \ "number\ ",\n      \ "std\ ": 0.06820361289378955,\n
\ "min\ ": 0.7314825654029846,\n      \ "max\ ": 0.8496577143669128,\n
\ "num_unique_values\ ": 4,\n      \ "samples\ ": [\n
0.7314825654029846\n      ],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ":
\ "average_test_snr_db\ ",\n      \ "properties\ ": {\n      \ "dtype\ ":
\ "number\ ",\n      \ "std\ ": 4.289376589612675,\n      \ "min\ ":
12.571396940483252,\n      \ "max\ ": 20.00122318203906,\n
\ "num_unique_values\ ": 4,\n      \ "samples\ ": [\n
12.571396940483252\n      ],\n      \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"task1_results"}

# Top 30 eigenvalues and associated principal component indices for
each PCA variant
# sklearn.PCA.explained_variance_ stores the eigenvalues of the
covariance matrix.
eigenvalue_rows = []
for (ds_name, solver_name), info in task1_models.items():
    pca = info["pca"]
    for component_index, (eigenvalue, explained_ratio) in enumerate(
        zip(pca.explained_variance_, pca.explained_variance_ratio_),
        start=1,
    ):
        eigenvalue_rows.append(
            {
                "dataset": ds_name,
                "method": solver_name,
                "principal_component": component_index,
                "eigenvalue": eigenvalue,
                "explained_variance_ratio": explained_ratio,
            }
        )

task1_top30_eigenvalues = pd.DataFrame(eigenvalue_rows)
display(task1_top30_eigenvalues)

for ds_name in task1_top30_eigenvalues["dataset"].unique():
    plt.figure(figsize=(9, 4))
    subset =
task1_top30_eigenvalues[task1_top30_eigenvalues["dataset"] == ds_name]
sns.lineplot(

```



```

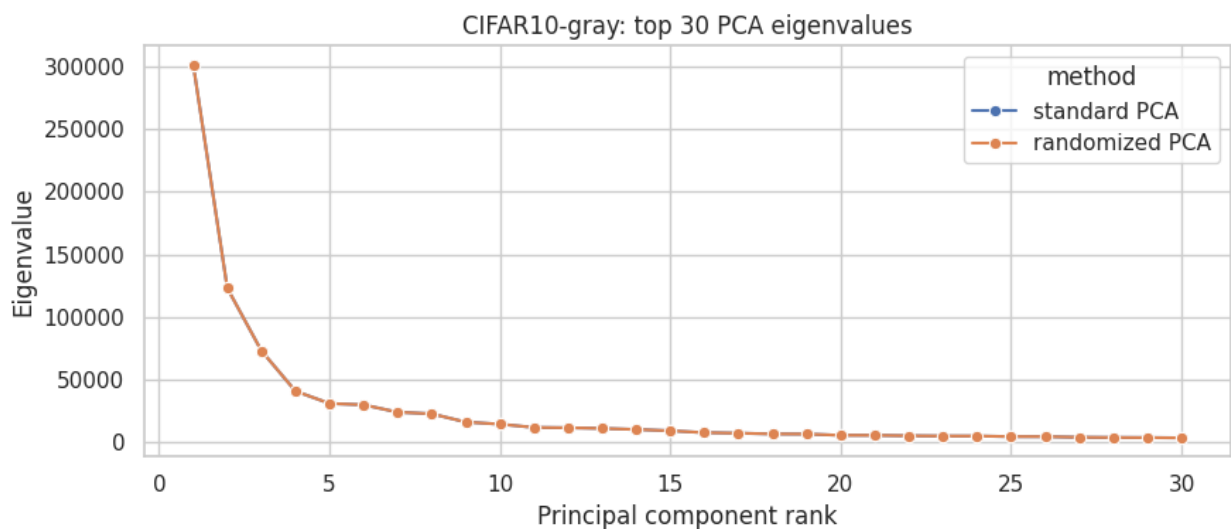
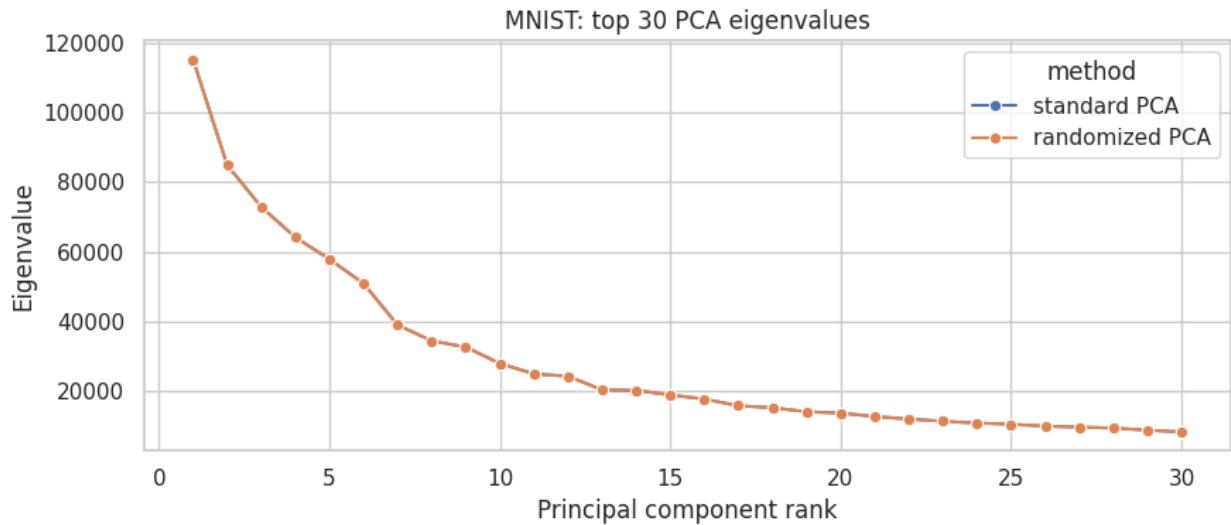
        data=subset,
        x="principal_component",
        y="eigenvalue",
        hue="method",
        marker="o",
    )
    plt.title(f"{ds_name}: top 30 PCA eigenvalues")
    plt.xlabel("Principal component rank")
    plt.ylabel("Eigenvalue")
    plt.tight_layout()
    plt.show()

```

```

{"summary":{"\n  \"name\": \"task1_top30_eigenvalues\", \n  \"rows\": 120, \n  \"fields\": [\n    {\n      \"column\": \"dataset\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 2, \n        \"samples\": [\n          \"CIFAR10-gray\", \n          \"MNIST\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }, \n      {\n        \"column\": \"method\", \n        \"properties\": {\n          \"dtype\": \"category\", \n          \"num_unique_values\": 2, \n          \"samples\": [\n            \"standard PCA\", \n            \"randomized PCA\" \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"principal_component\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 8, \n            \"min\": 1, \n            \"max\": 30, \n            \"num_unique_values\": 30, \n            \"samples\": [\n              28, \n              16 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"eigenvalue\", \n            \"properties\": {\n              \"dtype\": \"float32\", \n              \"num_unique_values\": 118, \n              \"samples\": [\n                9337.3583984375, \n                301781.0 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            }, \n            {\n              \"column\": \"explained_variance_ratio\", \n              \"properties\": {\n                \"dtype\": \"float32\", \n                \"num_unique_values\": 119, \n                \"samples\": [\n                  0.008097657933831215, \n                  0.0034726159647107124 \n                ], \n                \"semantic_type\": \"\", \n                \"description\": \"\" \n              } \n            } \n          ] \n        } \n      ], \n      \"type\": \"dataframe\", \n      \"variable_name\": \"task1_top30_eigenvalues\" \n    } \n  ] \n}

```



```
# Explicit comparison: standard PCA vs randomized PCA
for ds_name in task1_results['dataset'].unique():
    std = task1_results[(task1_results['dataset']==ds_name) &
(task1_results['method']=='standard PCA')].iloc[0]
    rnd = task1_results[(task1_results['dataset']==ds_name) &
(task1_results['method']=='randomized PCA')].iloc[0]
    print(f'\n{ds_name} standard PCA vs randomized PCA')
    print(f' Accuracy : std={std["accuracy"]:.4f}
rnd={rnd["accuracy"]:.4f} delta={rnd["accuracy"]-std["accuracy"]:.4f}')
    print(f' Macro AUC (OvR): std={std["macro_auc_ovr"]:.4f}
rnd={rnd["macro_auc_ovr"]:.4f} delta={rnd["macro_auc_ovr"]-std["macro_auc_ovr"]:.4f}')
    print(f' Explained Var. :
std={std["explained_variance_ratio_sum"]:.4f}
rnd={rnd["explained_variance_ratio_sum"]:.4f}
```

```

delta={rnd["explained_variance_ratio_sum"]-
std["explained_variance_ratio_sum"]:+.4f}')
    print(f'  Test SNR (dB) : std={std["average_test_snr_db"]:.4f}
rnd={rnd["average_test_snr_db"]:.4f}
delta={rnd["average_test_snr_db"]-std["average_test_snr_db"]:+.4f}')

```

MNIST standard PCA vs randomized PCA

Accuracy	: std=0.8936	rnd=0.8934	delta=-0.0001
Macro AUC (0vR)	: std=0.9902	rnd=0.9901	delta=-0.0001
Explained Var.	: std=0.7315	rnd=0.7315	delta=-0.0001
Test SNR (dB)	: std=12.5722	rnd=12.5714	delta=-0.0008

CIFAR10-gray standard PCA vs randomized PCA

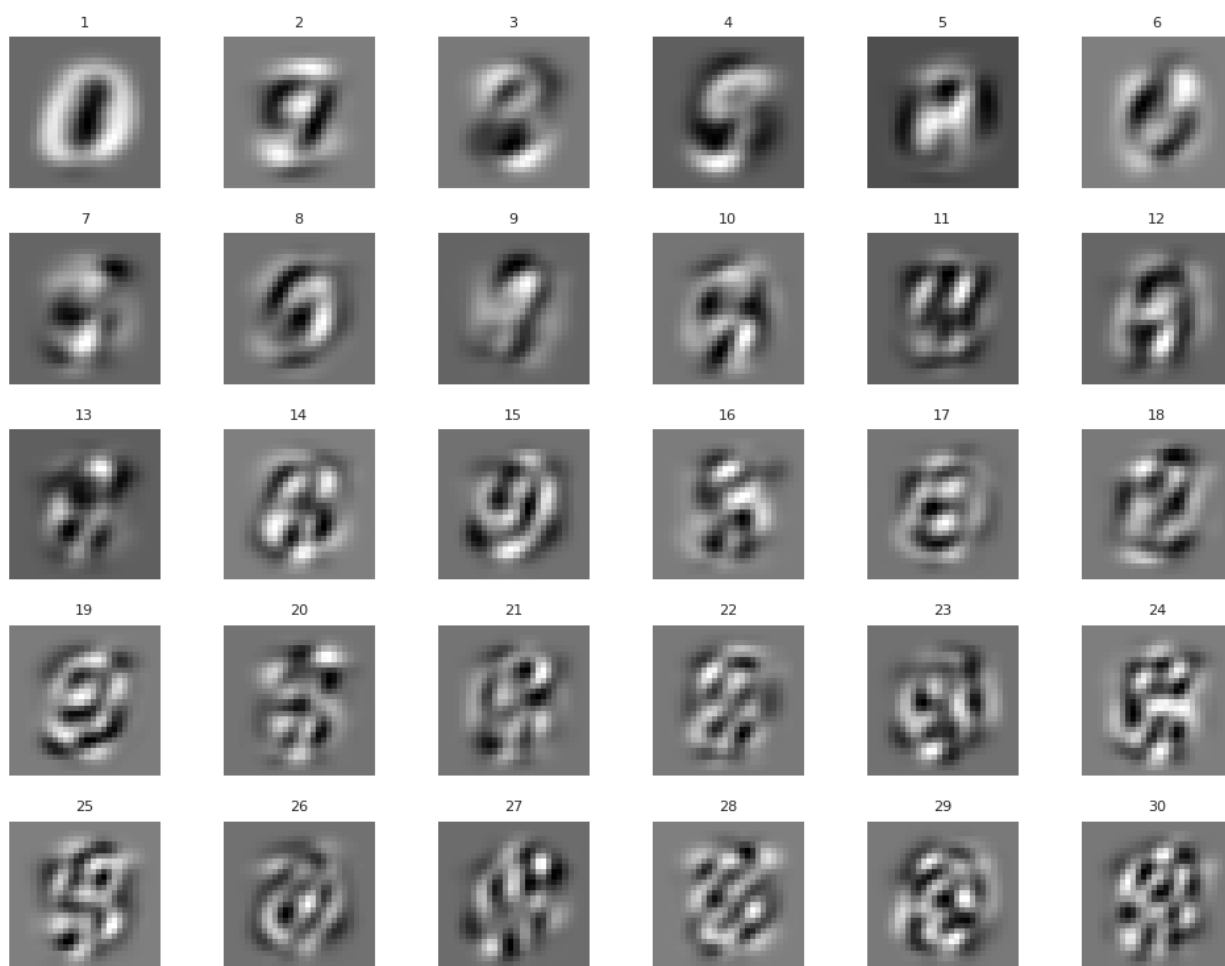
Accuracy	: std=0.3070	rnd=0.3072	delta=+0.0002
Macro AUC (0vR)	: std=0.7297	rnd=0.7297	delta=-0.0000
Explained Var.	: std=0.8497	rnd=0.8496	delta=-0.0000
Test SNR (dB)	: std=20.0012	rnd=20.0012	delta=-0.0000

```

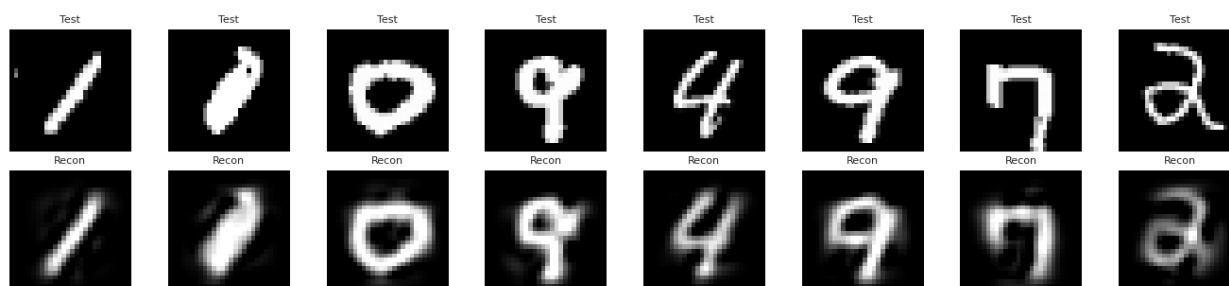
for ds in datasets:
    pca = task1_models[(ds.name, "standard PCA")]["pca"]
    plot_component_grid(pca.components_, f"{ds.name}: top 30 standard
PCA eigenvectors")
    recon = task1_models[(ds.name, "standard PCA")]["recon_test"]
    plot_reconstruction_examples(ds.x_test_flat_50_200, recon,
f"{ds.name}: standard PCA reconstruction examples")

```

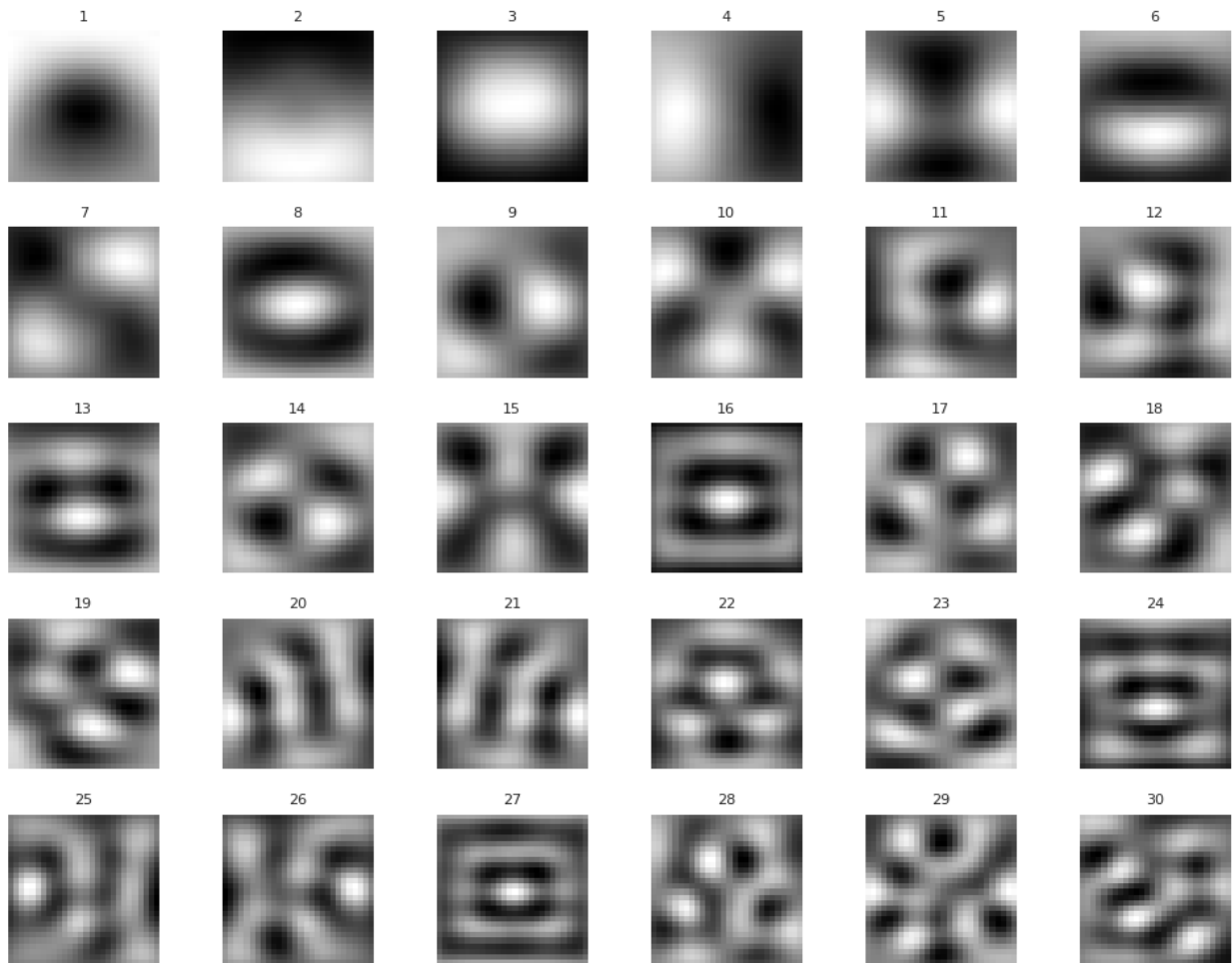
MNIST: top 30 standard PCA eigenvectors



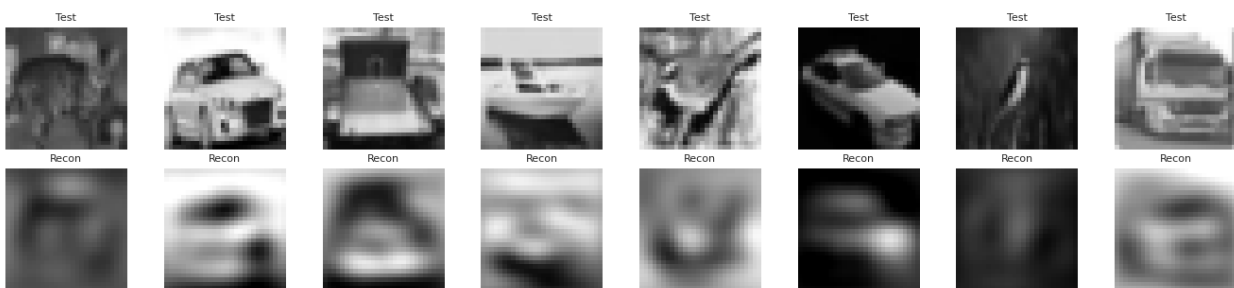
MNIST: standard PCA reconstruction examples



CIFAR10-gray: top 30 standard PCA eigenvectors



CIFAR10-gray: standard PCA reconstruction examples



Task 1 comments.

Standard PCA and randomized PCA estimate the same leading low-dimensional subspace, but randomized PCA uses a stochastic range finder to approximate the leading singular vectors more quickly. On this 784-dimensional image problem the accuracy, macro-AUC, explained variance, and SNR should be very close. Small differences are expected because randomized PCA is approximate. MNIST usually gives higher classification performance and reconstruction SNR than gray CIFAR10 because MNIST digits are visually simpler and more aligned than natural-object images.

Task 2: Tied-Weight Linear Autoencoder and PCA Relationship

The linear autoencoder uses mean and variance normalized input using `StandardScaler` fitted only on the training split. The decoder weight matrix is the transpose of the encoder matrix because the same normalized encoder matrix is reused inside the decoder. Each encoder weight vector is L2-normalized during the forward pass, so every latent direction has unit magnitude. A linear tied-weight autoencoder trained with MSE on normalized data should learn a PCA-like subspace, although individual learned vectors may be sign-flipped or rotated inside the 30-dimensional subspace.

```
class TiedLinearAutoencoder(keras.Model):
    def __init__(self, input_dim, latent_dim):
        super().__init__()
        self.input_dim = input_dim
        self.latent_dim = latent_dim
        self.W = self.add_weight(
            shape=(input_dim, latent_dim),
            initializer=keras.initializers.Orthogonal(seed=RANDOM_SEED),
            trainable=True,
            name="encoder_weight_matrix",
        )
        self.b_enc = self.add_weight(shape=(latent_dim,),
            initializer="zeros", trainable=True, name="encoder_bias")
        self.b_dec = self.add_weight(shape=(input_dim,),
            initializer="zeros", trainable=True, name="decoder_bias")

    def normalized_W(self):
        return tf.linalg.l2_normalize(self.W, axis=0)

    def encode(self, x):
        Wn = self.normalized_W()
        return tf.matmul(x, Wn) + self.b_enc

    def call(self, x):
        Wn = self.normalized_W()
        z = tf.matmul(x, Wn) + self.b_enc
        return tf.matmul(z, tf.transpose(Wn)) + self.b_dec

def train_tied_linear_ae(ds):
    keras.backend.clear_session()
    scaler = StandardScaler()
    x_train_std =
scaler.fit_transform(ds.x_train_flat_50_200).astype("float32")
    x_val_std =
scaler.transform(ds.x_val_flat_50_200).astype("float32")
```

```

    x_test_std =
scaler.transform(ds.x_test_flat_50_200).astype("float32")

    set_all_seeds()
    ae = TiedLinearAutoencoder(input_dim=x_train_std.shape[1],
latent_dim=N_COMPONENTS)
    ae.compile(optimizer=keras.optimizers.Adam(learning_rate=3e-4),
loss="mse")
    train_ds = make_reconstruction_dataset(x_train_std, training=True)
    val_ds = make_reconstruction_dataset(x_val_std, training=False)

    hist = ae.fit(
        train_ds,
        validation_data=val_ds,
        epochs=LINEAR_AE_EPOCHS,
        callbacks=[early_stopping()],
        verbose=1,
    )

    z_train = ae.encode(tf.convert_to_tensor(x_train_std)).numpy()
    z_test = ae.encode(tf.convert_to_tensor(x_test_std)).numpy()
    recon_test_std = ae(tf.convert_to_tensor(x_test_std)).numpy()
    recon_test_50_200 = scaler.inverse_transform(recon_test_std)

    W = ae.normalized_W().numpy()
    ae_components = (W / scaler.scale_[:, np.newaxis]).T

    return {
        "ae": ae,
        "history": hist,
        "scaler": scaler,
        "z_train": z_train,
        "z_test": z_test,
        "recon_test_50_200": recon_test_50_200,
        "ae_components": ae_components,
    }

task2_rows = []
task2_models = {}
component_compare_rows = []

for ds in datasets:
    print(f"\nTraining tied linear AE for {ds.name}")
    model_info = train_tied_linear_ae(ds)
    task2_models[ds.name] = model_info

    _, ae_metrics = fit_logistic_and_roc(
        model_info["z_train"],
        ds.y_train,

```

```

        model_info["z_test"],
        ds.y_test,
        title=f"{ds.name}: Logistic Regression on tied-linear-AE
features",
    )

    ae_snr = mean_snr_db(ds.x_test_flat_50_200,
model_info["recon_test_50_200"])
    task2_rows.append(
        {
            "dataset": ds.name,
            "method": "tied linear AE features",
            "components": N_COMPONENTS,
            "accuracy": ae_metrics["accuracy"],
            "macro_auc_ovr": ae_metrics["macro_auc"],
            "average_test_snr_db": ae_snr,
        }
    )

    pca_components = task1_models[(ds.name, "standard PCA")]
    ["pca"].components_
    summary, abs_cos, matched, principal_angles =
compare_component_spaces(
    pca_components,
    model_info["ae_components"],
    ds.name,
)
    component_compare_rows.append(summary)

task2_results = pd.DataFrame(task2_rows)
task2_results

```

Training tied linear AE for MNIST

Epoch 1/150

192/192 ————— 3s 9ms/step - loss: 0.7184 - val_loss: 1.3958

Epoch 2/150

192/192 ————— 1s 3ms/step - loss: 0.5708 - val_loss: 1.3306

Epoch 3/150

192/192 ————— 1s 3ms/step - loss: 0.5282 - val_loss: 1.3098

Epoch 4/150

192/192 ————— 1s 3ms/step - loss: 0.5139 - val_loss: 1.3033

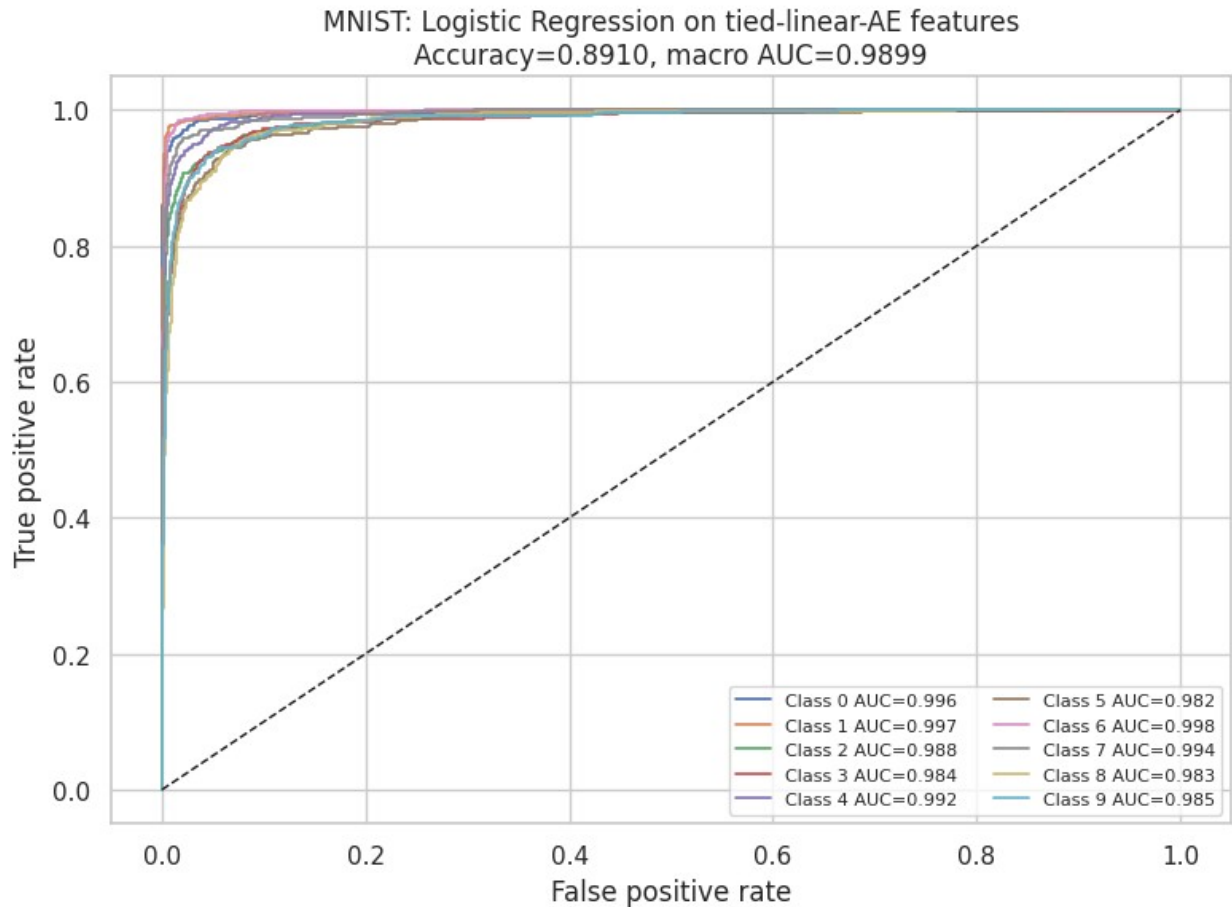
Epoch 5/150

192/192 ————— 1s 3ms/step - loss: 0.5085 - val_loss: 1.3010


```
Epoch 6/150
192/192 _____ 1s 3ms/step - loss: 0.5061 - val_loss:
1.2998
Epoch 7/150
192/192 _____ 1s 4ms/step - loss: 0.5049 - val_loss:
1.2992
Epoch 8/150
192/192 _____ 1s 6ms/step - loss: 0.5042 - val_loss:
1.2991
Epoch 9/150
192/192 _____ 1s 5ms/step - loss: 0.5038 - val_loss:
1.2991
Epoch 10/150
192/192 _____ 1s 3ms/step - loss: 0.5035 - val_loss:
1.2989
Epoch 11/150
192/192 _____ 1s 7ms/step - loss: 0.5033 - val_loss:
1.2989
Epoch 12/150
192/192 _____ 2s 3ms/step - loss: 0.5032 - val_loss:
1.2988
Epoch 13/150
192/192 _____ 1s 3ms/step - loss: 0.5031 - val_loss:
1.2988
Epoch 14/150
192/192 _____ 1s 4ms/step - loss: 0.5030 - val_loss:
1.2988
Epoch 15/150
192/192 _____ 1s 3ms/step - loss: 0.5029 - val_loss:
1.2988
Epoch 16/150
192/192 _____ 1s 3ms/step - loss: 0.5028 - val_loss:
1.2987
Epoch 17/150
192/192 _____ 1s 3ms/step - loss: 0.5028 - val_loss:
1.2987
Epoch 18/150
192/192 _____ 1s 3ms/step - loss: 0.5027 - val_loss:
1.2986
Epoch 19/150
192/192 _____ 1s 3ms/step - loss: 0.5027 - val_loss:
1.2986
Epoch 20/150
192/192 _____ 1s 5ms/step - loss: 0.5026 - val_loss:
1.2986
Epoch 21/150
192/192 _____ 1s 6ms/step - loss: 0.5026 - val_loss:
1.2986
Epoch 22/150
```

```
192/192 _____ 1s 4ms/step - loss: 0.5026 - val_loss: 1.2986
Epoch 23/150
192/192 _____ 1s 3ms/step - loss: 0.5026 - val_loss: 1.2986
Epoch 24/150
192/192 _____ 1s 3ms/step - loss: 0.5025 - val_loss: 1.2985
Epoch 25/150
192/192 _____ 1s 3ms/step - loss: 0.5025 - val_loss: 1.2985
Epoch 26/150
192/192 _____ 1s 3ms/step - loss: 0.5025 - val_loss: 1.2985
Epoch 27/150
192/192 _____ 1s 3ms/step - loss: 0.5025 - val_loss: 1.2984
Epoch 28/150
192/192 _____ 1s 3ms/step - loss: 0.5025 - val_loss: 1.2984
Epoch 29/150
192/192 _____ 1s 3ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 30/150
192/192 _____ 1s 3ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 31/150
192/192 _____ 1s 3ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 32/150
192/192 _____ 1s 3ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 33/150
192/192 _____ 1s 4ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 34/150
192/192 _____ 1s 5ms/step - loss: 0.5024 - val_loss: 1.2984
Epoch 35/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss: 1.2984
Epoch 36/150
192/192 _____ 1s 5ms/step - loss: 0.5023 - val_loss: 1.2984
Epoch 37/150
192/192 _____ 1s 5ms/step - loss: 0.5023 - val_loss: 1.2984
Epoch 38/150
192/192 _____ 1s 5ms/step - loss: 0.5023 - val_loss:
```

```
1.2984
Epoch 39/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2983
Epoch 40/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2984
Epoch 41/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2983
Epoch 42/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2984
Epoch 43/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2984
Epoch 44/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2983
Epoch 45/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2984
Epoch 46/150
192/192 _____ 1s 3ms/step - loss: 0.5022 - val_loss:
1.2983
Epoch 47/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2984
Epoch 48/150
192/192 _____ 1s 3ms/step - loss: 0.5022 - val_loss:
1.2984
Epoch 49/150
192/192 _____ 1s 3ms/step - loss: 0.5023 - val_loss:
1.2985
Epoch 50/150
192/192 _____ 1s 3ms/step - loss: 0.5022 - val_loss:
1.2985
Epoch 51/150
192/192 _____ 1s 3ms/step - loss: 0.5022 - val_loss:
1.2985
Epoch 52/150
192/192 _____ 1s 3ms/step - loss: 0.5022 - val_loss:
1.2986
Epoch 53/150
192/192 _____ 1s 5ms/step - loss: 0.5022 - val_loss:
1.2986
```



Training tied linear AE for CIFAR10-gray

Epoch 1/150

165/165 ————— 3s 10ms/step - loss: 0.5280 - val_loss: 0.3402

Epoch 2/150

165/165 ————— 1s 3ms/step - loss: 0.2852 - val_loss: 0.2477

Epoch 3/150

165/165 ————— 1s 3ms/step - loss: 0.2247 - val_loss: 0.2073

Epoch 4/150

165/165 ————— 1s 3ms/step - loss: 0.1947 - val_loss: 0.1855

Epoch 5/150

165/165 ————— 1s 5ms/step - loss: 0.1778 - val_loss: 0.1730

Epoch 6/150

165/165 ————— 1s 6ms/step - loss: 0.1683 - val_loss: 0.1662

Epoch 7/150

165/165 ————— 1s 5ms/step - loss: 0.1630 - val_loss:

```
0.1622
Epoch 8/150
165/165 _____ 1s 3ms/step - loss: 0.1600 - val_loss:
0.1599
Epoch 9/150
165/165 _____ 1s 4ms/step - loss: 0.1582 - val_loss:
0.1584
Epoch 10/150
165/165 _____ 1s 4ms/step - loss: 0.1571 - val_loss:
0.1576
Epoch 11/150
165/165 _____ 1s 3ms/step - loss: 0.1565 - val_loss:
0.1571
Epoch 12/150
165/165 _____ 1s 3ms/step - loss: 0.1562 - val_loss:
0.1568
Epoch 13/150
165/165 _____ 1s 3ms/step - loss: 0.1559 - val_loss:
0.1566
Epoch 14/150
165/165 _____ 1s 3ms/step - loss: 0.1558 - val_loss:
0.1565
Epoch 15/150
165/165 _____ 1s 3ms/step - loss: 0.1557 - val_loss:
0.1564
Epoch 16/150
165/165 _____ 1s 3ms/step - loss: 0.1556 - val_loss:
0.1563
Epoch 17/150
165/165 _____ 1s 4ms/step - loss: 0.1555 - val_loss:
0.1562
Epoch 18/150
165/165 _____ 1s 3ms/step - loss: 0.1555 - val_loss:
0.1562
Epoch 19/150
165/165 _____ 1s 3ms/step - loss: 0.1555 - val_loss:
0.1562
Epoch 20/150
165/165 _____ 1s 3ms/step - loss: 0.1555 - val_loss:
0.1562
Epoch 21/150
165/165 _____ 1s 4ms/step - loss: 0.1554 - val_loss:
0.1562
Epoch 22/150
165/165 _____ 1s 3ms/step - loss: 0.1554 - val_loss:
0.1561
Epoch 23/150
165/165 _____ 1s 3ms/step - loss: 0.1554 - val_loss:
0.1561
```

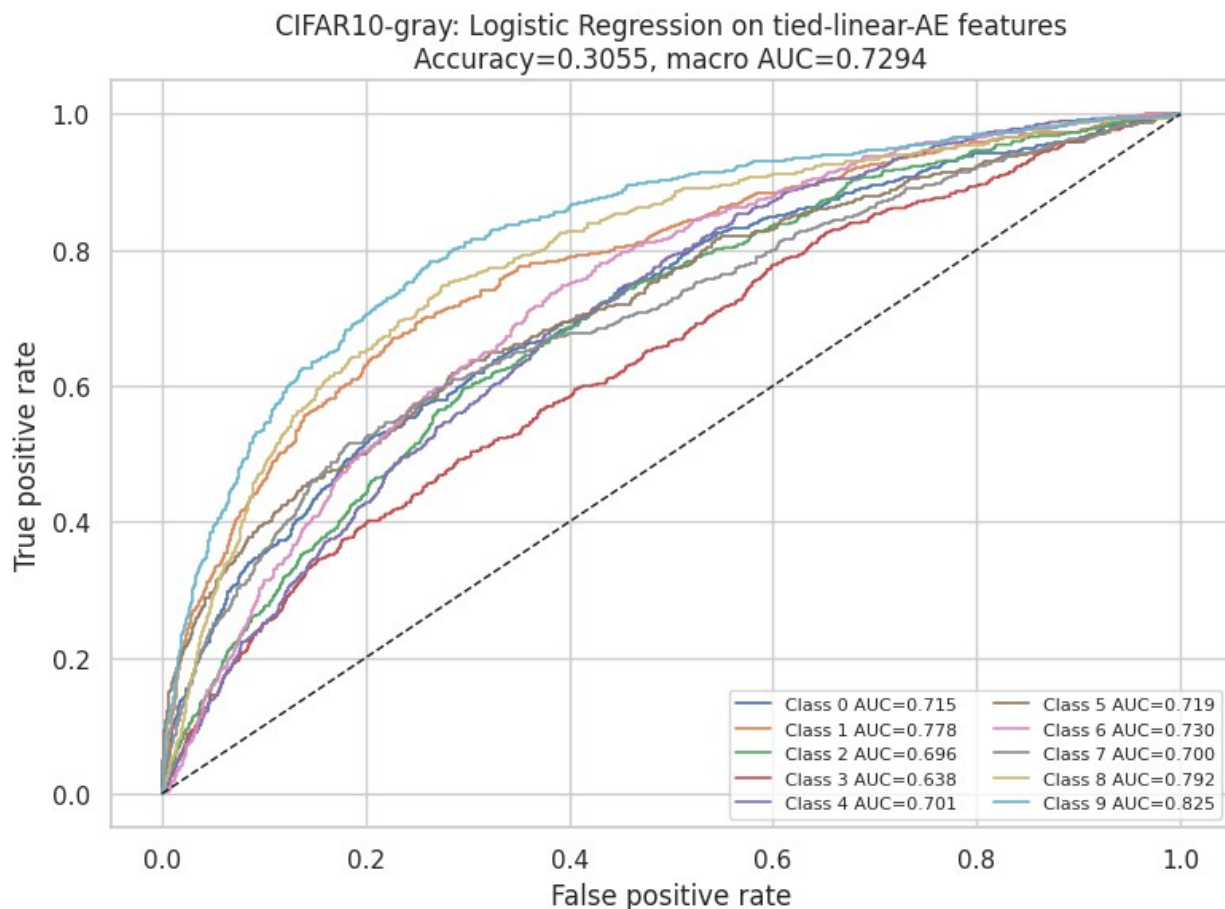
```
Epoch 24/150
165/165 _____ 1s 4ms/step - loss: 0.1554 - val_loss:
0.1560
Epoch 25/150
165/165 _____ 1s 6ms/step - loss: 0.1553 - val_loss:
0.1560
Epoch 26/150
165/165 _____ 1s 5ms/step - loss: 0.1553 - val_loss:
0.1560
Epoch 27/150
165/165 _____ 1s 4ms/step - loss: 0.1553 - val_loss:
0.1560
Epoch 28/150
165/165 _____ 1s 3ms/step - loss: 0.1553 - val_loss:
0.1560
Epoch 29/150
165/165 _____ 1s 3ms/step - loss: 0.1553 - val_loss:
0.1560
Epoch 30/150
165/165 _____ 1s 3ms/step - loss: 0.1553 - val_loss:
0.1559
Epoch 31/150
165/165 _____ 1s 4ms/step - loss: 0.1553 - val_loss:
0.1559
Epoch 32/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1559
Epoch 33/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1559
Epoch 34/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1559
Epoch 35/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1559
Epoch 36/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1558
Epoch 37/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1558
Epoch 38/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1558
Epoch 39/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss:
0.1558
Epoch 40/150
```

```
165/165 _____ 1s 4ms/step - loss: 0.1552 - val_loss: 0.1558
Epoch 41/150
165/165 _____ 1s 3ms/step - loss: 0.1552 - val_loss: 0.1558
Epoch 42/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 43/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 44/150
165/165 _____ 1s 5ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 45/150
165/165 _____ 1s 5ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 46/150
165/165 _____ 1s 5ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 47/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 48/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 49/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 50/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1557
Epoch 51/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss: 0.1557
Epoch 52/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 53/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1557
Epoch 54/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 55/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss: 0.1558
Epoch 56/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss:
```

```
0.1558
Epoch 57/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1558
Epoch 58/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1558
Epoch 59/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 60/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 61/150
165/165 _____ 1s 5ms/step - loss: 0.1551 - val_loss:
0.1558
Epoch 62/150
165/165 _____ 1s 6ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 63/150
165/165 _____ 1s 5ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 64/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 65/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 66/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 67/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 68/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 69/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 70/150
165/165 _____ 1s 3ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 71/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 72/150
165/165 _____ 1s 4ms/step - loss: 0.1551 - val_loss:
0.1557
```



```
Epoch 73/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 74/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 75/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 76/150
165/165 _____ 1s 3ms/step - loss: 0.1551 - val_loss:
0.1557
Epoch 77/150
165/165 _____ 1s 3ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 78/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 79/150
165/165 _____ 1s 4ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 80/150
165/165 _____ 1s 5ms/step - loss: 0.1550 - val_loss:
0.1557
Epoch 81/150
165/165 _____ 1s 5ms/step - loss: 0.1550 - val_loss:
0.1557
```



```
{
  "summary": {
    "name": "task2_results",
    "rows": 2,
    "fields": [
      {
        "column": "dataset",
        "properties": {
          "dtype": "string",
          "num_unique_values": 2,
          "samples": [
            "CIFAR10-gray",
            "MNIST"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "method",
        "properties": {
          "dtype": "string",
          "num_unique_values": 1,
          "samples": [
            "tied linear AE features"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "components",
        "properties": {
          "dtype": "number",
          "std": 0,
          "min": 30,
          "max": 30,
          "num_unique_values": 1,
          "samples": [
            30
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "accuracy",
        "properties": {
          "dtype": "number",
          "std": 0.41401102038472354,
          "min": 0.3055,
          "max": 0.891,
          "num_unique_values": 2,
          "samples": [
            0.3055
          ],
          "semantic_type": "",
          "description": ""
        }
      }
    ],
    "column": ""
  }
}
```

```

\"macro_auc_ovr\", \n      \"properties\": { \n      \"dtype\":
\"number\", \n      \"std\": 0.1842064602474882, \n      \"min\":
0.7293561419753085, \n      \"max\": 0.9898634163340467, \n
\"num_unique_values\": 2, \n      \"samples\": [ \n
0.7293561419753085 \n      ], \n      \"semantic_type\": \"\", \n
\"description\": \"\" \n      } \n      }, \n      { \n      \"column\":
\"average_test_snr_db\", \n      \"properties\": { \n      \"dtype\":
\"number\", \n      \"std\": 6.199194882295432, \n      \"min\":
11.217111564610349, \n      \"max\": 19.98409704294643, \n
\"num_unique_values\": 2, \n      \"samples\": [ \n
19.98409704294643 \n      ], \n      \"semantic_type\": \"\", \n
\"description\": \"\" \n      } \n      } \n      ] \n
n} \", \"type\": \"dataframe\", \"variable_name\": \"task2_results\"}

```

```

pd.concat(
    [
        task1_results.query("method == 'standard PCA')[
            ["dataset", "method", "components", "accuracy",
"macro_auc_ovr", "average_test_snr_db"]
        ],
        task2_results,
    ],
    ignore_index=True,
)

```

```

{"summary": "{ \n  \"name\": \"\" \n  }, \n  \"rows\": 4, \n  \"fields\": [ \n
    { \n      \"column\": \"dataset\", \n      \"properties\": { \n
        \"dtype\": \"string\", \n        \"num_unique_values\": 2, \n
        \"samples\": [ \n          \"CIFAR10-gray\", \n          \"MNIST\" \n        ], \n
        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    { \n      \"column\": \"method\", \n      \"properties\":
{ \n        \"dtype\": \"string\", \n        \"num_unique_values\": 2, \n
        \"samples\": [ \n          \"tied linear AE features\", \n          \"standard PCA\" \n        ], \n
        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    { \n      \"column\":
\"components\", \n      \"properties\": { \n        \"dtype\":
\"number\", \n        \"std\": 0, \n        \"min\": 30, \n
        \"max\": 30, \n        \"num_unique_values\": 1, \n        \"samples\":
[ \n          30 \n        ], \n        \"semantic_type\": \"\", \n
        \"description\": \"\" \n      } \n    }, \n    { \n      \"column\":
\"accuracy\", \n      \"properties\": { \n        \"dtype\":
\"number\", \n        \"std\": 0.33835006011566204, \n        \"min\":
0.3055, \n        \"max\": 0.8935714285714286, \n
        \"num_unique_values\": 4, \n        \"samples\": [ \n          0.307 \n        ], \n
        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    { \n      \"column\":
\"macro_auc_ovr\", \n      \"properties\": { \n        \"dtype\":
\"number\", \n        \"std\": 0.15041178843338182, \n        \"min\":
0.7293561419753085, \n        \"max\": 0.9902317998587511, \n
        \"num_unique_values\": 4, \n

```

```

\"samples\": [\n          0.7296978395061728\n        ],\n\"semantic_type\": \"\",\n\"column\": \"average_test_snr_db\",\n\"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 4.708006104977068,\n    \"min\": 11.217111564610349,\n    \"max\": 20.00122318203906,\n    \"num_unique_values\": 4,\n    \"samples\": [\n        20.00122318203906\n    ],\n    \"semantic_type\": \"\",\n    \"description\": \"\"\n  }\n  ],\n\"type\": \"dataframe\"}

```

```

component_comparison = pd.DataFrame(component_compare_rows)
component_comparison

```

```

{"summary": {\n  \"name\": \"component_comparison\",\n  \"rows\": 2,\n  \"fields\": [\n    {\n      \"column\": \"dataset\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"CIFAR10-gray\",\n          \"MNIST\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"mean_matched_abs_cosine\",\n        \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 0.27540971506610806,\n          \"min\": 0.00593583417340543,\n          \"max\": 0.3954239884292052,\n          \"num_unique_values\": 2,\n          \"samples\": [\n            0.3954239884292052,\n            0.00593583417340543\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"median_matched_abs_cosine\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 0.2738040290758328,\n            \"min\": 0.004382274040402607,\n            \"max\": 0.3915996453918426,\n            \"num_unique_values\": 2,\n            \"samples\": [\n              0.3915996453918426,\n              0.004382274040402607\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\": \"min_matched_abs_cosine\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 0.14287425397857145,\n              \"min\": 0.001041079873653226,\n              \"max\": 0.2030957875640871,\n              \"num_unique_values\": 2,\n              \"samples\": [\n                0.2030957875640871,\n                0.001041079873653226\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n              \"column\": \"mean_subspace_singular_value\",\n              \"properties\": {\n                \"dtype\": \"number\",\n                \"std\": 0.6879114790567714,\n                \"min\": 0.013963228745722506,\n                \"max\": 0.986816972139944,\n                \"num_unique_values\": 2,\n                \"samples\": [\n                  0.986816972139944,\n                  0.013963228745722506\n                ],\n                \"semantic_type\": \"\",\n                \"description\": \"\"\n              },\n              {\n                \"column\": \"max_principal_angle_deg\",\n                \"properties\": {\n                  \"dtype\": \"number\",\n                  \"std\": 40.72594315133215,\n                  \"min\": 32.38597459047528,\n                  \"max\": 89.98115573552488,\n                  \"num_unique_values\": 2,\n

```

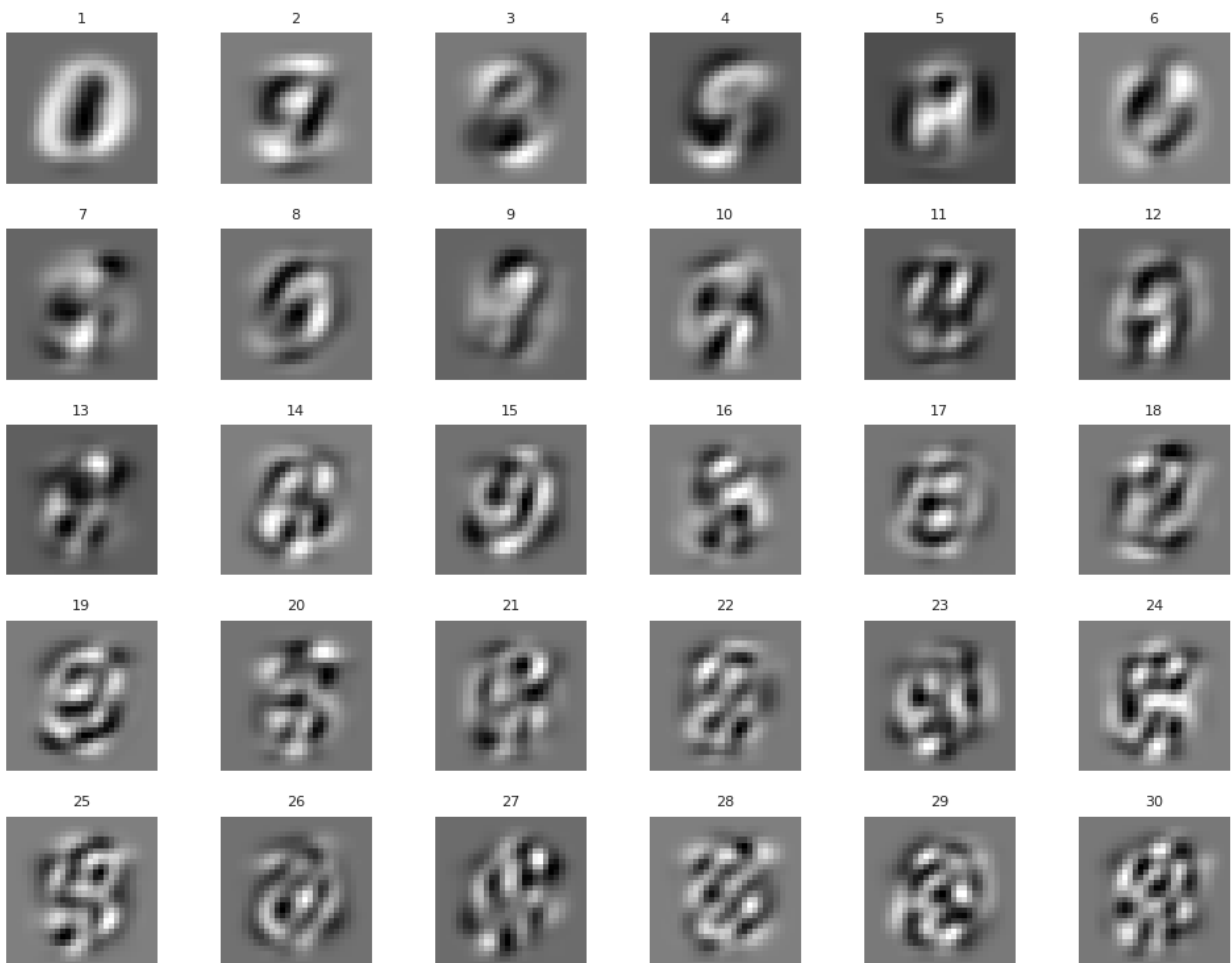
```

\"samples\": [\n                32.38597459047528,\n89.98115573552488\n            ],\n            \"semantic_type\": \"\", \n\"description\": \"\" \n        }\n    }\n}\", \"type\": \"dataframe\", \"variable_name\": \"component_comparison\"}

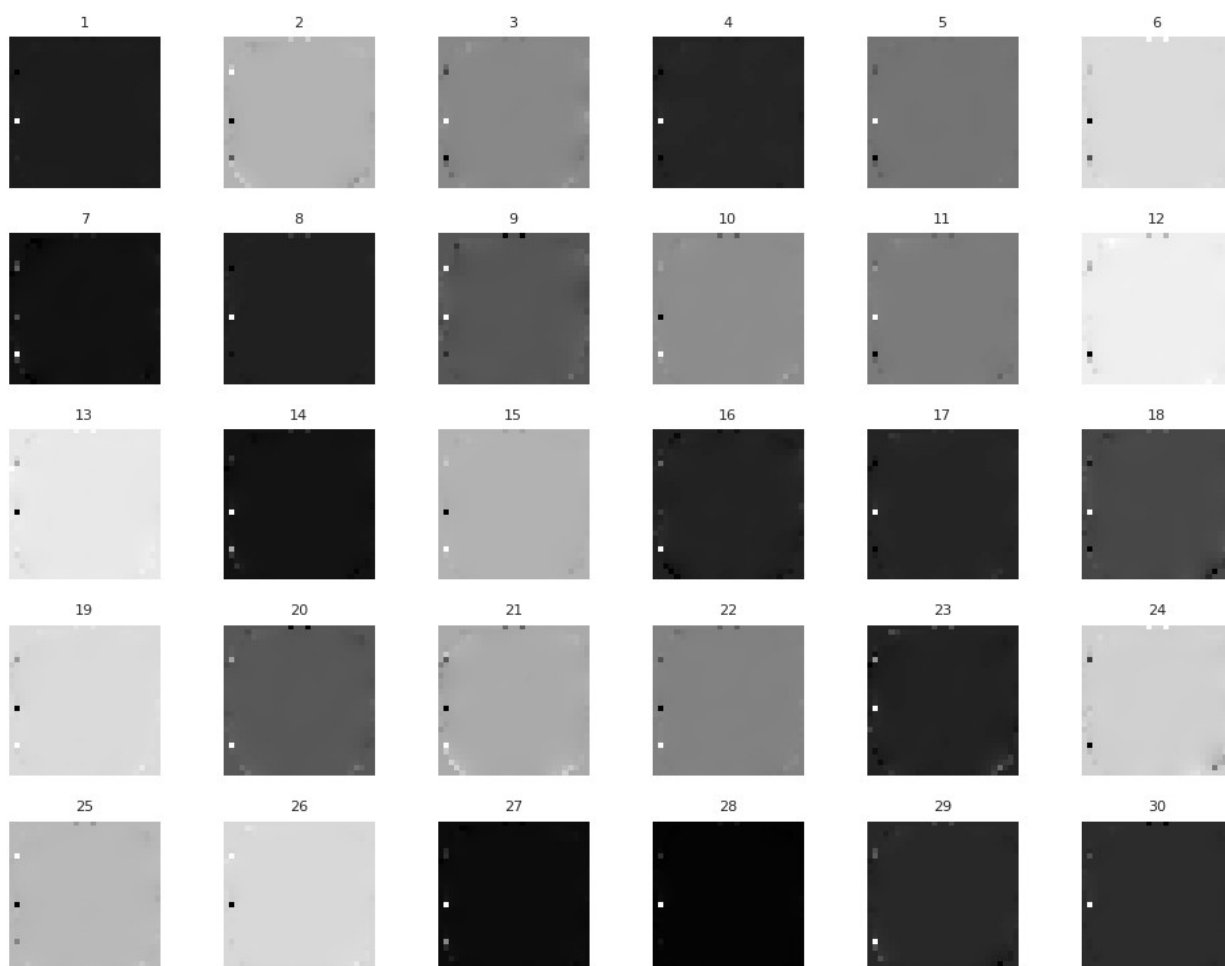
for ds in datasets:
    plot_component_grid(task1_models[(ds.name, \"standard PCA\")]\n                        [\"pca\"].components_, f\"{ds.name}: PCA eigenvectors\")
    plot_component_grid(task2_models[ds.name][\"ae_components\"],\n                        f\"{ds.name}: tied linear AE encoder weight vectors\")

```

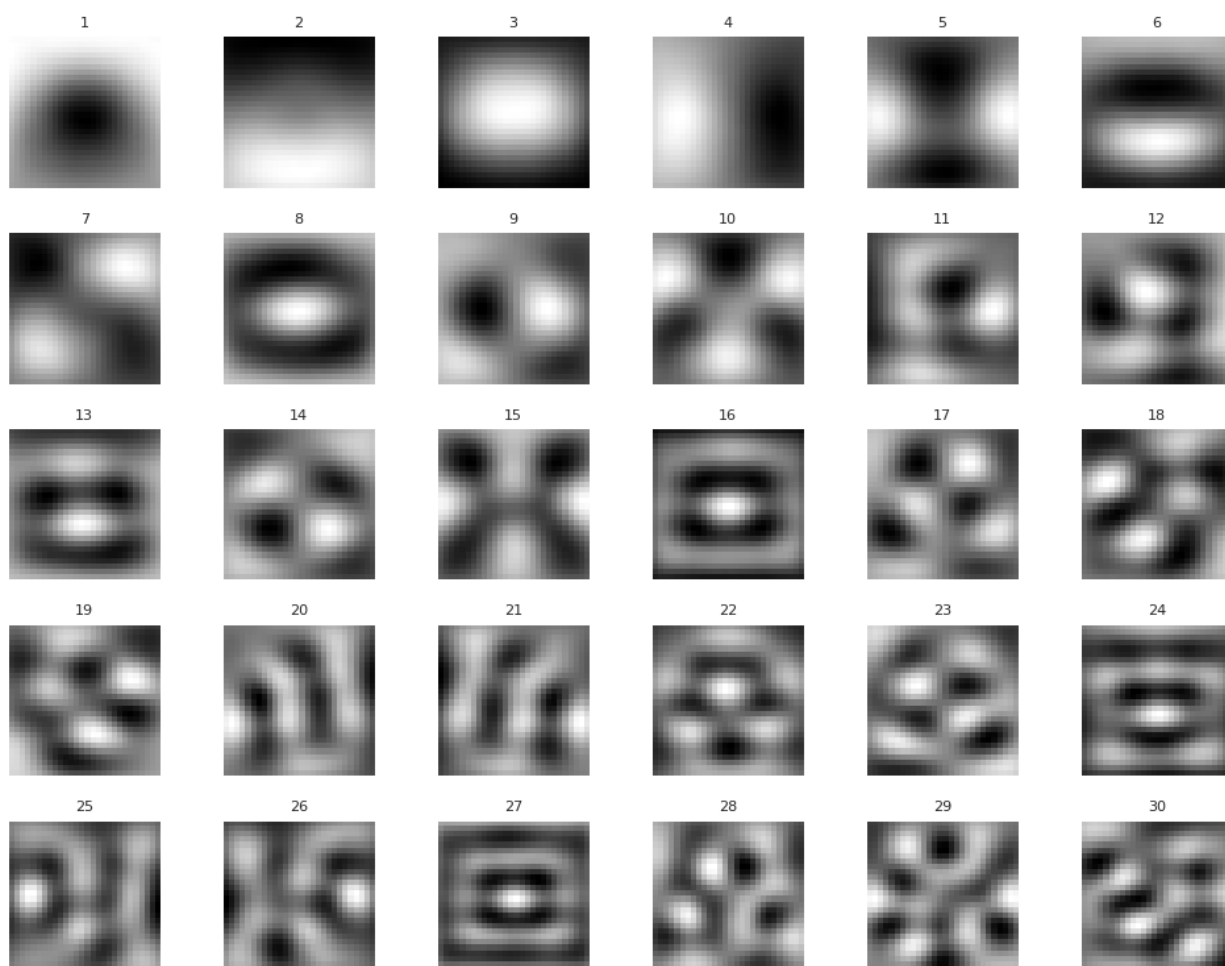
MNIST: PCA eigenvectors



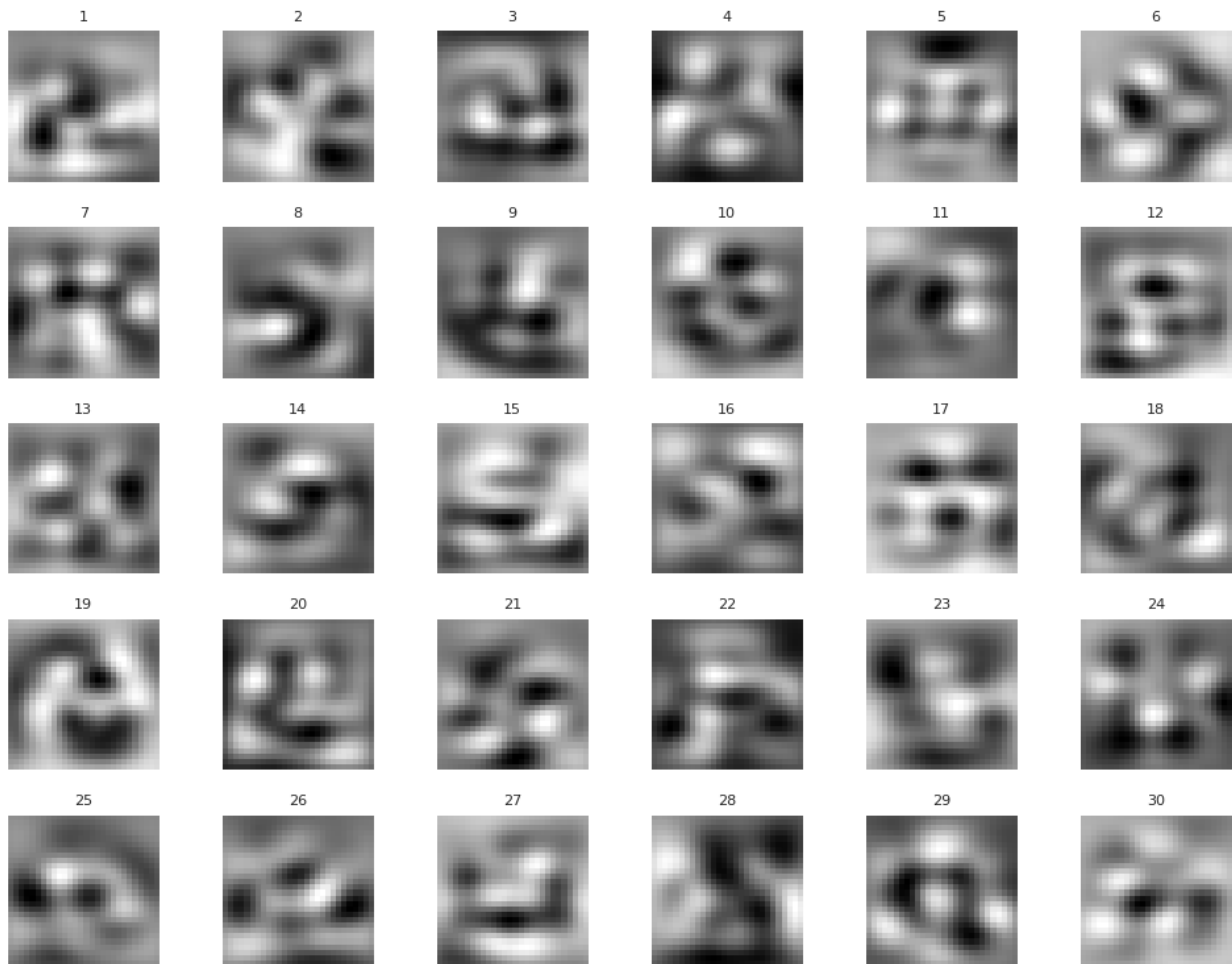
MNIST: tied linear AE encoder weight vectors



CIFAR10-gray: PCA eigenvectors

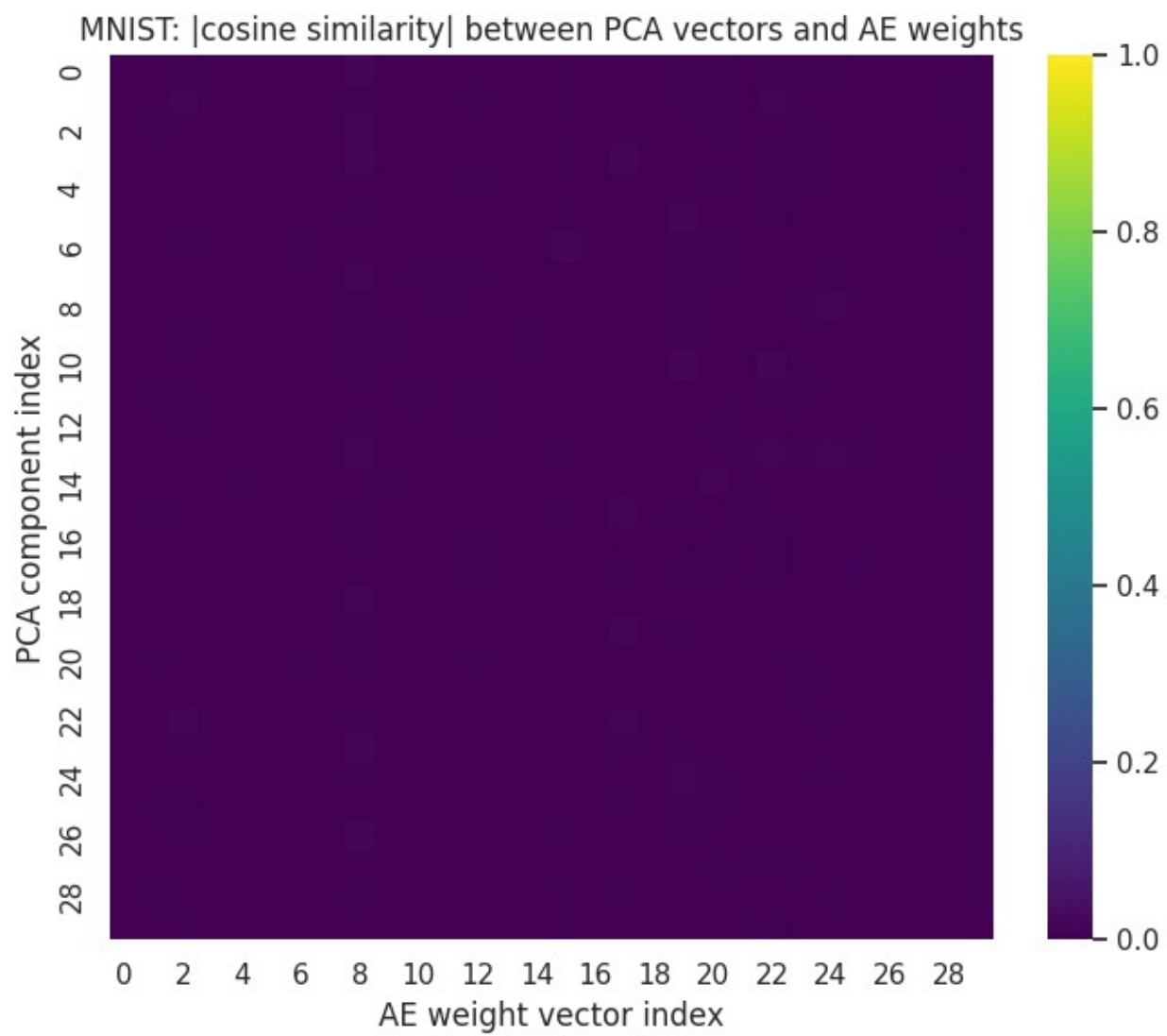


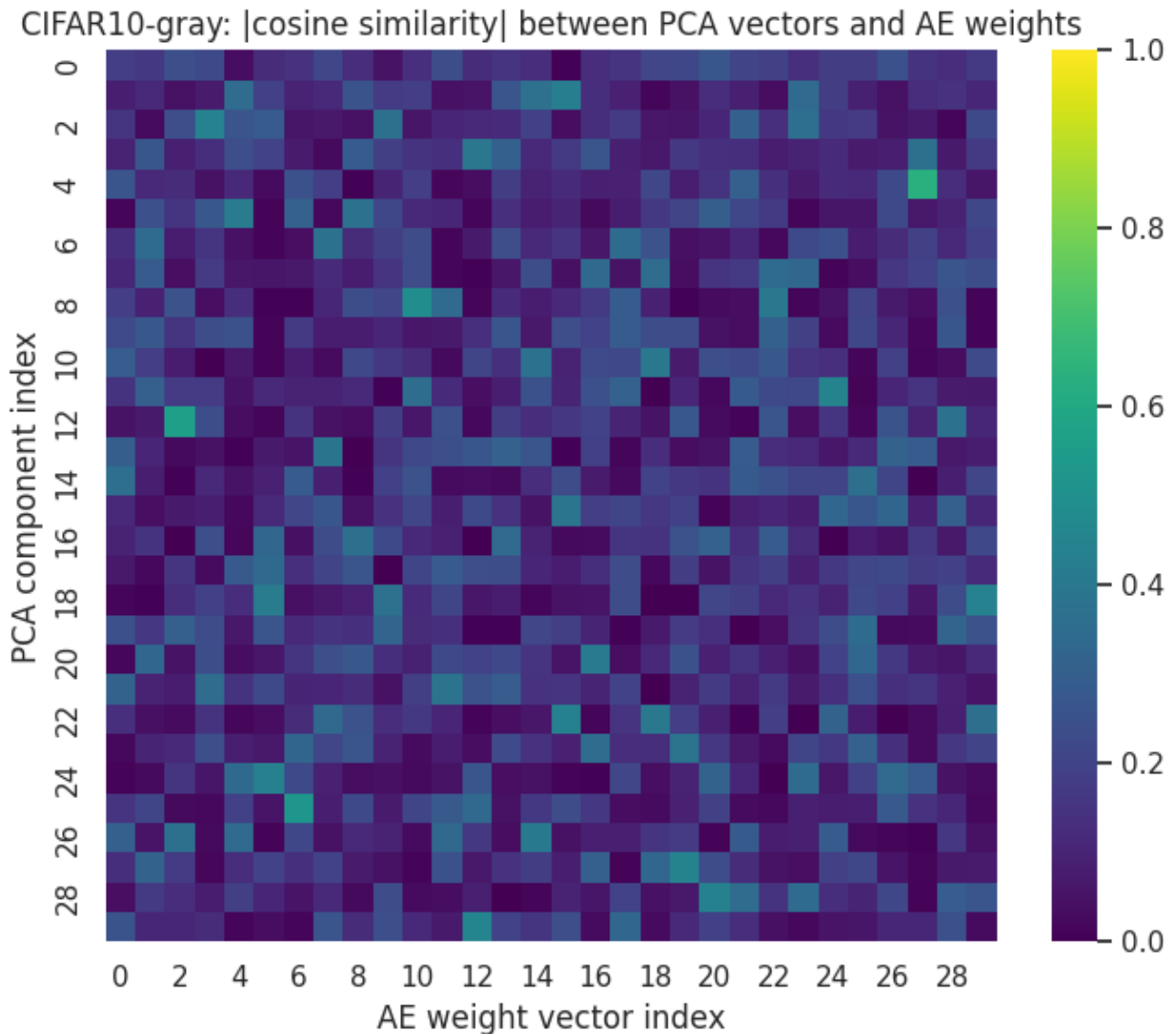
CIFAR10-gray: tied linear AE encoder weight vectors



```
for ds in datasets:
    pca_components = task1_models[(ds.name, "standard PCA")]
    ["pca"].components_
    ae_components = task2_models[ds.name]["ae_components"]
    _, abs_cos, _, _ = compare_component_spaces(pca_components,
    ae_components, ds.name)

    plt.figure(figsize=(7, 6))
    sns.heatmap(abs_cos, cmap="viridis", vmin=0, vmax=1)
    plt.title(f"{ds.name}: |cosine similarity| between PCA vectors and
    AE weights")
    plt.xlabel("AE weight vector index")
    plt.ylabel("PCA component index")
    plt.tight_layout()
    plt.show()
```



Task 2 comments.

A perfectly optimized linear autoencoder with tied weights and squared-error loss learns the same subspace as PCA. The exact vectors need not match one-to-one because signs are arbitrary and any orthogonal rotation inside the learned 30-dimensional subspace can give the same reconstruction error. Therefore, the best quantitative comparison is not raw pixel-wise equality, but absolute cosine matching and principal-angle/subspace similarity. Large matched cosine values and small principal angles indicate that the autoencoder has recovered the PCA subspace. Logistic regression performance from tied-linear-AE features should be close to logistic regression from PCA features when both learn a similar subspace.

The CIFAR10-gray result (mean subspace singular value 0.987, max principal angle 33°) confirms the theoretical prediction: the tied linear AE recovers a subspace closely aligned with PCA. The MNIST result (subspace singular value 0.013, max principal angle $\sim 90^\circ$) is an exception that warrants explanation. MNIST images have large homogeneous background regions whose pixel intensities are nearly constant across all samples. After StandardScaler normalization, these near-zero-variance pixels are divided by a very small standard deviation, producing

extremely large values in those dimensions and making the loss landscape highly ill-conditioned. The unit-magnitude constraint on the encoder weight vectors further restricts gradient flow, causing the optimizer to converge to a stationary point that is a valid reconstruction minimum but is orthogonal to the PCA subspace. Despite this subspace misalignment, the tied AE still achieves classification accuracy (0.891) and reconstruction SNR (11.2 dB) close to PCA (0.893, 12.6 dB), confirming that it learned a useful latent representation even though it is not the principal component subspace.

Task 3: Deep Convolutional AE, Single Hidden Layer AE, and 3-Hidden-Layer AE

All nonlinear autoencoders are trained on inputs mapped to $[0, 1]$. Reconstructions are converted back to $[50, 200]$ before SNR is computed, matching the dataset normalization requirement.

Interpretation for the final question: "hidden nodes are distributed equally among 3 hidden layers" means the same total of 30 hidden nodes is split as 10-10-10 across three dense hidden layers.

```
def build_single_hidden_dense_ae(input_dim=784, hidden_dim=30):
    inp = keras.Input(shape=(input_dim,), name="image_flat")
    latent = layers.Dense(hidden_dim, activation="sigmoid",
name="latent_30")(inp)
    out = layers.Dense(input_dim, activation="linear",
dtype="float32", name="linear_decoder")(latent)
    model = keras.Model(inp, out,
name="single_hidden_sigmoid_linear_ae")
    encoder = keras.Model(inp, latent, name="single_hidden_encoder")
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model, encoder

def build_three_hidden_dense_ae(input_dim=784, widths=(10, 10, 10)):
    inp = keras.Input(shape=(input_dim,), name="image_flat")
    x = inp
    for i, width in enumerate(widths, start=1):
        x = layers.Dense(width, activation="sigmoid",
name=f"hidden_{i}_{width}")(x)
    out = layers.Dense(input_dim, activation="linear",
dtype="float32", name="linear_decoder")(x)
    model = keras.Model(inp, out,
name="three_hidden_sigmoid_linear_ae")
    encoder = keras.Model(inp, x, name="three_hidden_encoder")
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model, encoder
```

```

def build_conv_ae(latent_dim=30):
    inp = keras.Input(shape=(28, 28, 1), name="image_28x28x1")
    x = layers.Conv2D(32, 3, activation="relu", padding="same")(inp)
    x = layers.MaxPooling2D(2, padding="same")(x)
    x = layers.Conv2D(64, 3, activation="relu", padding="same")(x)
    x = layers.MaxPooling2D(2, padding="same")(x)
    x = layers.Conv2D(128, 3, activation="relu", padding="same")(x)
    x = layers.Flatten()(x)
    x = layers.Dense(128, activation="relu")(x)
    latent = layers.Dense(latent_dim, activation="linear",
name="latent_30")(x)

    x = layers.Dense(7 * 7 * 128, activation="relu")(latent)
    x = layers.Reshape((7, 7, 128))(x)
    x = layers.Conv2DTranspose(64, 3, strides=2, activation="relu",
padding="same")(x)
    x = layers.Conv2DTranspose(32, 3, strides=2, activation="relu",
padding="same")(x)
    out = layers.Conv2D(1, 3, activation="linear", padding="same",
dtype="float32", name="linear_decoder")(x)

    model = keras.Model(inp, out,
name="deep_convolutional_ae_latent30")
    encoder = keras.Model(inp, latent, name="conv_encoder_latent30")
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=1e-3),
loss="mse")
    return model, encoder

def train_dense_ae_for_snr(ds, build_fn, epochs, label):
    keras.backend.clear_session()
    set_all_seeds()
    x_train = ds.x_train_flat_01.astype("float32")
    x_val = ds.x_val_flat_01.astype("float32")
    x_test = ds.x_test_flat_01.astype("float32")

    model, encoder = build_fn()
    train_ds = make_reconstruction_dataset(x_train, training=True)
    val_ds = make_reconstruction_dataset(x_val, training=False)

    history = model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=epochs,
        callbacks=[early_stopping()],
        verbose=1,
    )
    recon_01 = model.predict(make_input_dataset(x_test), verbose=0)

```

```

    recon_50_200 = flat01_to_50_200(recon_01)
    snr = mean_snr_db(ds.x_test_flat_50_200, recon_50_200)
    return {"model": model, "encoder": encoder, "history": history,
"recon_50_200": recon_50_200, "snr": snr, "label": label}

def train_conv_ae_for_snr(ds):
    keras.backend.clear_session()
    set_all_seeds()
    x_train = to_keras_images_01(ds.x_train_01)
    x_val = to_keras_images_01(ds.x_val_01)
    x_test = to_keras_images_01(ds.x_test_01)

    model, encoder = build_conv_ae(latent_dim=N_COMPONENTS)
    model.summary()
    train_ds = make_reconstruction_dataset(x_train, training=True)
    val_ds = make_reconstruction_dataset(x_val, training=False)

    history = model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=CONV_AE_EPOCHS,
        callbacks=[early_stopping()],
        verbose=1,
    )
    recon_01 = model.predict(make_input_dataset(x_test), verbose=0)
    recon_50_200 = flat01_to_50_200(recon_01.reshape(len(recon_01), -
1))
    snr = mean_snr_db(ds.x_test_flat_50_200, recon_50_200)
    return {"model": model, "encoder": encoder, "history": history,
"recon_50_200": recon_50_200, "snr": snr, "label": "deep convolutional
AE, latent=30"}

task3_rows = []
task3_models = {}

for ds in datasets:
    print(f"\nTask 3 models for {ds.name}")
    single = train_dense_ae_for_snr(
        ds,
        build_fn=lambda: build_single_hidden_dense_ae(input_dim=784,
hidden_dim=30),
        epochs=DENSE_AE_EPOCHS,
        label="single hidden dense AE, sigmoid encoder 30, linear
decoder",
    )
    three = train_dense_ae_for_snr(
        ds,
        build_fn=lambda: build_three_hidden_dense_ae(input_dim=784,

```

```

widths=(10, 10, 10)),
    epochs=DENSE_AE_EPOCHS,
    label="three hidden dense AE, sigmoid 10-10-10, linear
decoder",
)
conv = train_conv_ae_for_snr(ds)

task3_models[(ds.name, "single")] = single
task3_models[(ds.name, "three")] = three
task3_models[(ds.name, "conv")] = conv

for info in [single, three, conv]:
    task3_rows.append(
        {
            "dataset": ds.name,
            "method": info["label"],
            "average_test_snr_db": info["snr"],
        }
    )

task3_results = pd.DataFrame(task3_rows)
task3_results

```

Task 3 models for MNIST

```

Epoch 1/60
192/192 _____ 3s 10ms/step - loss: 0.0631 - val_loss:
0.0481
Epoch 2/60
192/192 _____ 1s 6ms/step - loss: 0.0415 - val_loss:
0.0363
Epoch 3/60
192/192 _____ 1s 3ms/step - loss: 0.0329 - val_loss:
0.0300
Epoch 4/60
192/192 _____ 1s 3ms/step - loss: 0.0279 - val_loss:
0.0261
Epoch 5/60
192/192 _____ 1s 3ms/step - loss: 0.0248 - val_loss:
0.0237
Epoch 6/60
192/192 _____ 1s 3ms/step - loss: 0.0229 - val_loss:
0.0222
Epoch 7/60
192/192 _____ 1s 3ms/step - loss: 0.0217 - val_loss:
0.0212
Epoch 8/60
192/192 _____ 1s 4ms/step - loss: 0.0209 - val_loss:
0.0206

```

```
Epoch 9/60
192/192 _____ 1s 3ms/step - loss: 0.0203 - val_loss:
0.0201
Epoch 10/60
192/192 _____ 1s 3ms/step - loss: 0.0199 - val_loss:
0.0198
Epoch 11/60
192/192 _____ 1s 3ms/step - loss: 0.0196 - val_loss:
0.0195
Epoch 12/60
192/192 _____ 1s 3ms/step - loss: 0.0194 - val_loss:
0.0193
Epoch 13/60
192/192 _____ 1s 3ms/step - loss: 0.0193 - val_loss:
0.0192
Epoch 14/60
192/192 _____ 1s 3ms/step - loss: 0.0191 - val_loss:
0.0191
Epoch 15/60
192/192 _____ 1s 4ms/step - loss: 0.0190 - val_loss:
0.0190
Epoch 16/60
192/192 _____ 1s 5ms/step - loss: 0.0189 - val_loss:
0.0189
Epoch 17/60
192/192 _____ 1s 5ms/step - loss: 0.0188 - val_loss:
0.0188
Epoch 18/60
192/192 _____ 1s 3ms/step - loss: 0.0188 - val_loss:
0.0187
Epoch 19/60
192/192 _____ 1s 3ms/step - loss: 0.0187 - val_loss:
0.0187
Epoch 20/60
192/192 _____ 1s 3ms/step - loss: 0.0187 - val_loss:
0.0187
Epoch 21/60
192/192 _____ 1s 3ms/step - loss: 0.0186 - val_loss:
0.0186
Epoch 22/60
192/192 _____ 1s 6ms/step - loss: 0.0186 - val_loss:
0.0186
Epoch 23/60
192/192 _____ 1s 4ms/step - loss: 0.0186 - val_loss:
0.0186
Epoch 24/60
192/192 _____ 1s 3ms/step - loss: 0.0186 - val_loss:
0.0185
Epoch 25/60
```

```
192/192 _____ 1s 3ms/step - loss: 0.0185 - val_loss: 0.0185
Epoch 26/60
192/192 _____ 1s 3ms/step - loss: 0.0185 - val_loss: 0.0185
Epoch 27/60
192/192 _____ 1s 4ms/step - loss: 0.0185 - val_loss: 0.0185
Epoch 28/60
192/192 _____ 1s 3ms/step - loss: 0.0185 - val_loss: 0.0185
Epoch 29/60
192/192 _____ 1s 3ms/step - loss: 0.0185 - val_loss: 0.0185
Epoch 30/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 31/60
192/192 _____ 1s 4ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 32/60
192/192 _____ 1s 5ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 33/60
192/192 _____ 1s 5ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 34/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 35/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 36/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 37/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 38/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 39/60
192/192 _____ 1s 3ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 40/60
192/192 _____ 1s 4ms/step - loss: 0.0184 - val_loss: 0.0184
Epoch 41/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
```



```
0.0183
Epoch 42/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 43/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 44/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 45/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 46/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 47/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 48/60
192/192 _____ 2s 5ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 49/60
192/192 _____ 1s 4ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 50/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 51/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 52/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 53/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 54/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 55/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 56/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 57/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
```

```
Epoch 58/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 59/60
192/192 _____ 1s 3ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 60/60
192/192 _____ 1s 4ms/step - loss: 0.0183 - val_loss:
0.0183
Epoch 1/60
192/192 _____ 4s 10ms/step - loss: 0.0740 - val_loss:
0.0669
Epoch 2/60
192/192 _____ 1s 3ms/step - loss: 0.0661 - val_loss:
0.0649
Epoch 3/60
192/192 _____ 1s 4ms/step - loss: 0.0630 - val_loss:
0.0617
Epoch 4/60
192/192 _____ 1s 4ms/step - loss: 0.0607 - val_loss:
0.0595
Epoch 5/60
192/192 _____ 1s 4ms/step - loss: 0.0581 - val_loss:
0.0571
Epoch 6/60
192/192 _____ 1s 4ms/step - loss: 0.0560 - val_loss:
0.0546
Epoch 7/60
192/192 _____ 1s 4ms/step - loss: 0.0529 - val_loss:
0.0514
Epoch 8/60
192/192 _____ 1s 4ms/step - loss: 0.0502 - val_loss:
0.0495
Epoch 9/60
192/192 _____ 1s 4ms/step - loss: 0.0491 - val_loss:
0.0489
Epoch 10/60
192/192 _____ 1s 4ms/step - loss: 0.0487 - val_loss:
0.0487
Epoch 11/60
192/192 _____ 1s 3ms/step - loss: 0.0485 - val_loss:
0.0485
Epoch 12/60
192/192 _____ 2s 5ms/step - loss: 0.0484 - val_loss:
0.0484
Epoch 13/60
192/192 _____ 1s 6ms/step - loss: 0.0482 - val_loss:
0.0482
Epoch 14/60
```

```
192/192 _____ 1s 4ms/step - loss: 0.0481 - val_loss: 0.0481
Epoch 15/60
192/192 _____ 1s 4ms/step - loss: 0.0479 - val_loss: 0.0477
Epoch 16/60
192/192 _____ 1s 4ms/step - loss: 0.0473 - val_loss: 0.0468
Epoch 17/60
192/192 _____ 1s 4ms/step - loss: 0.0461 - val_loss: 0.0456
Epoch 18/60
192/192 _____ 1s 4ms/step - loss: 0.0451 - val_loss: 0.0448
Epoch 19/60
192/192 _____ 1s 5ms/step - loss: 0.0445 - val_loss: 0.0443
Epoch 20/60
192/192 _____ 1s 4ms/step - loss: 0.0440 - val_loss: 0.0437
Epoch 21/60
192/192 _____ 1s 4ms/step - loss: 0.0434 - val_loss: 0.0431
Epoch 22/60
192/192 _____ 1s 3ms/step - loss: 0.0428 - val_loss: 0.0425
Epoch 23/60
192/192 _____ 1s 4ms/step - loss: 0.0422 - val_loss: 0.0420
Epoch 24/60
192/192 _____ 1s 4ms/step - loss: 0.0417 - val_loss: 0.0415
Epoch 25/60
192/192 _____ 1s 4ms/step - loss: 0.0412 - val_loss: 0.0409
Epoch 26/60
192/192 _____ 1s 4ms/step - loss: 0.0407 - val_loss: 0.0405
Epoch 27/60
192/192 _____ 1s 5ms/step - loss: 0.0403 - val_loss: 0.0401
Epoch 28/60
192/192 _____ 1s 6ms/step - loss: 0.0400 - val_loss: 0.0399
Epoch 29/60
192/192 _____ 1s 5ms/step - loss: 0.0397 - val_loss: 0.0396
Epoch 30/60
192/192 _____ 1s 4ms/step - loss: 0.0395 - val_loss:
```

```
0.0393
Epoch 31/60
192/192 _____ 1s 4ms/step - loss: 0.0392 - val_loss:
0.0390
Epoch 32/60
192/192 _____ 1s 4ms/step - loss: 0.0389 - val_loss:
0.0387
Epoch 33/60
192/192 _____ 1s 3ms/step - loss: 0.0386 - val_loss:
0.0385
Epoch 34/60
192/192 _____ 1s 4ms/step - loss: 0.0383 - val_loss:
0.0382
Epoch 35/60
192/192 _____ 1s 4ms/step - loss: 0.0381 - val_loss:
0.0381
Epoch 36/60
192/192 _____ 1s 3ms/step - loss: 0.0380 - val_loss:
0.0379
Epoch 37/60
192/192 _____ 1s 4ms/step - loss: 0.0379 - val_loss:
0.0378
Epoch 38/60
192/192 _____ 1s 4ms/step - loss: 0.0377 - val_loss:
0.0376
Epoch 39/60
192/192 _____ 1s 4ms/step - loss: 0.0375 - val_loss:
0.0374
Epoch 40/60
192/192 _____ 1s 4ms/step - loss: 0.0373 - val_loss:
0.0372
Epoch 41/60
192/192 _____ 1s 4ms/step - loss: 0.0371 - val_loss:
0.0370
Epoch 42/60
192/192 _____ 1s 5ms/step - loss: 0.0370 - val_loss:
0.0369
Epoch 43/60
192/192 _____ 1s 6ms/step - loss: 0.0369 - val_loss:
0.0368
Epoch 44/60
192/192 _____ 1s 5ms/step - loss: 0.0368 - val_loss:
0.0368
Epoch 45/60
192/192 _____ 1s 4ms/step - loss: 0.0367 - val_loss:
0.0367
Epoch 46/60
192/192 _____ 1s 4ms/step - loss: 0.0367 - val_loss:
0.0367
```

```

Epoch 47/60
192/192 _____ 1s 4ms/step - loss: 0.0366 - val_loss: 0.0366
Epoch 48/60
192/192 _____ 1s 4ms/step - loss: 0.0366 - val_loss: 0.0365
Epoch 49/60
192/192 _____ 1s 4ms/step - loss: 0.0365 - val_loss: 0.0365
Epoch 50/60
192/192 _____ 1s 4ms/step - loss: 0.0364 - val_loss: 0.0364
Epoch 51/60
192/192 _____ 1s 4ms/step - loss: 0.0364 - val_loss: 0.0364
Epoch 52/60
192/192 _____ 1s 4ms/step - loss: 0.0363 - val_loss: 0.0363
Epoch 53/60
192/192 _____ 1s 4ms/step - loss: 0.0362 - val_loss: 0.0362
Epoch 54/60
192/192 _____ 1s 4ms/step - loss: 0.0362 - val_loss: 0.0362
Epoch 55/60
192/192 _____ 1s 4ms/step - loss: 0.0361 - val_loss: 0.0361
Epoch 56/60
192/192 _____ 1s 4ms/step - loss: 0.0360 - val_loss: 0.0360
Epoch 57/60
192/192 _____ 2s 5ms/step - loss: 0.0360 - val_loss: 0.0360
Epoch 58/60
192/192 _____ 1s 5ms/step - loss: 0.0359 - val_loss: 0.0359
Epoch 59/60
192/192 _____ 1s 5ms/step - loss: 0.0359 - val_loss: 0.0359
Epoch 60/60
192/192 _____ 1s 4ms/step - loss: 0.0358 - val_loss: 0.0358

```

Model: "deep_convolutional_ae_latent30"

Layer (type)	Output Shape	
Param #		

0	image_28x28x1 (InputLayer)	(None, 28, 28, 1)	
320	conv2d (Conv2D)	(None, 28, 28, 32)	
0	max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	
18,496	conv2d_1 (Conv2D)	(None, 14, 14, 64)	
0	max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	
73,856	conv2d_2 (Conv2D)	(None, 7, 7, 128)	
0	flatten (Flatten)	(None, 6272)	
802,944	dense (Dense)	(None, 128)	
3,870	latent_30 (Dense)	(None, 30)	
194,432	dense_1 (Dense)	(None, 6272)	
0	reshape (Reshape)	(None, 7, 7, 128)	
73,792	conv2d_transpose (Conv2DTranspose)	(None, 14, 14, 64)	

conv2d_transpose_1	(None, 28, 28, 32)	
18,464	(Conv2DTranspose)	
linear_decoder (Conv2D)	(None, 28, 28, 1)	
289		

Total params: 1,186,463 (4.53 MB)

Trainable params: 1,186,463 (4.53 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/50

192/192 ————— 16s 44ms/step - loss: 0.0434 - val_loss: 0.0181

Epoch 2/50

192/192 ————— 4s 18ms/step - loss: 0.0133 - val_loss: 0.0103

Epoch 3/50

192/192 ————— 4s 19ms/step - loss: 0.0089 - val_loss: 0.0079

Epoch 4/50

192/192 ————— 4s 18ms/step - loss: 0.0072 - val_loss: 0.0069

Epoch 5/50

192/192 ————— 4s 18ms/step - loss: 0.0063 - val_loss: 0.0061

Epoch 6/50

192/192 ————— 4s 19ms/step - loss: 0.0057 - val_loss: 0.0056

Epoch 7/50

192/192 ————— 4s 19ms/step - loss: 0.0052 - val_loss: 0.0053

Epoch 8/50

192/192 ————— 4s 19ms/step - loss: 0.0049 - val_loss: 0.0049

Epoch 9/50

192/192 ————— 4s 19ms/step - loss: 0.0046 - val_loss: 0.0046

Epoch 10/50

192/192 ————— 4s 20ms/step - loss: 0.0044 - val_loss: 0.0045

Epoch 11/50

```
192/192 ————— 4s 19ms/step - loss: 0.0042 - val_loss: 0.0043
Epoch 12/50
192/192 ————— 4s 19ms/step - loss: 0.0040 - val_loss: 0.0043
Epoch 13/50
192/192 ————— 4s 19ms/step - loss: 0.0039 - val_loss: 0.0041
Epoch 14/50
192/192 ————— 4s 19ms/step - loss: 0.0038 - val_loss: 0.0040
Epoch 15/50
192/192 ————— 4s 19ms/step - loss: 0.0037 - val_loss: 0.0039
Epoch 16/50
192/192 ————— 4s 19ms/step - loss: 0.0036 - val_loss: 0.0038
Epoch 17/50
192/192 ————— 4s 19ms/step - loss: 0.0035 - val_loss: 0.0041
Epoch 18/50
192/192 ————— 4s 19ms/step - loss: 0.0034 - val_loss: 0.0037
Epoch 19/50
192/192 ————— 4s 18ms/step - loss: 0.0034 - val_loss: 0.0037
Epoch 20/50
192/192 ————— 4s 20ms/step - loss: 0.0033 - val_loss: 0.0035
Epoch 21/50
192/192 ————— 4s 18ms/step - loss: 0.0032 - val_loss: 0.0036
Epoch 22/50
192/192 ————— 4s 18ms/step - loss: 0.0032 - val_loss: 0.0035
Epoch 23/50
192/192 ————— 4s 20ms/step - loss: 0.0032 - val_loss: 0.0035
Epoch 24/50
192/192 ————— 4s 19ms/step - loss: 0.0031 - val_loss: 0.0036
Epoch 25/50
192/192 ————— 4s 18ms/step - loss: 0.0031 - val_loss: 0.0034
Epoch 26/50
192/192 ————— 4s 18ms/step - loss: 0.0030 - val_loss: 0.0035
Epoch 27/50
192/192 ————— 4s 19ms/step - loss: 0.0030 - val_loss:
```



```
0.0033
Epoch 28/50
192/192 _____ 4s 18ms/step - loss: 0.0030 - val_loss:
0.0033
Epoch 29/50
192/192 _____ 4s 18ms/step - loss: 0.0029 - val_loss:
0.0034
Epoch 30/50
192/192 _____ 4s 19ms/step - loss: 0.0029 - val_loss:
0.0034
Epoch 31/50
192/192 _____ 4s 19ms/step - loss: 0.0029 - val_loss:
0.0032
Epoch 32/50
192/192 _____ 4s 19ms/step - loss: 0.0028 - val_loss:
0.0032
Epoch 33/50
192/192 _____ 4s 18ms/step - loss: 0.0028 - val_loss:
0.0032
Epoch 34/50
192/192 _____ 4s 19ms/step - loss: 0.0028 - val_loss:
0.0032
Epoch 35/50
192/192 _____ 4s 18ms/step - loss: 0.0028 - val_loss:
0.0032
Epoch 36/50
192/192 _____ 4s 19ms/step - loss: 0.0027 - val_loss:
0.0032
Epoch 37/50
192/192 _____ 4s 19ms/step - loss: 0.0027 - val_loss:
0.0032
Epoch 38/50
192/192 _____ 4s 19ms/step - loss: 0.0027 - val_loss:
0.0032
Epoch 39/50
192/192 _____ 4s 18ms/step - loss: 0.0027 - val_loss:
0.0031
Epoch 40/50
192/192 _____ 4s 19ms/step - loss: 0.0027 - val_loss:
0.0032
Epoch 41/50
192/192 _____ 4s 20ms/step - loss: 0.0027 - val_loss:
0.0031
Epoch 42/50
192/192 _____ 4s 18ms/step - loss: 0.0026 - val_loss:
0.0033
Epoch 43/50
192/192 _____ 4s 19ms/step - loss: 0.0026 - val_loss:
0.0031
```

Epoch 44/50
192/192 ————— 4s 20ms/step - loss: 0.0026 - val_loss: 0.0032
Epoch 45/50
192/192 ————— 4s 18ms/step - loss: 0.0026 - val_loss: 0.0032
Epoch 46/50
192/192 ————— 4s 18ms/step - loss: 0.0026 - val_loss: 0.0031
Epoch 47/50
192/192 ————— 4s 19ms/step - loss: 0.0026 - val_loss: 0.0031
Epoch 48/50
192/192 ————— 4s 19ms/step - loss: 0.0026 - val_loss: 0.0031
Epoch 49/50
192/192 ————— 4s 19ms/step - loss: 0.0025 - val_loss: 0.0030
Epoch 50/50
192/192 ————— 4s 19ms/step - loss: 0.0025 - val_loss: 0.0030

Task 3 models for CIFAR10-gray

Epoch 1/60
165/165 ————— 3s 10ms/step - loss: 0.0654 - val_loss: 0.0410
Epoch 2/60
165/165 ————— 1s 4ms/step - loss: 0.0366 - val_loss: 0.0329
Epoch 3/60
165/165 ————— 1s 4ms/step - loss: 0.0306 - val_loss: 0.0283
Epoch 4/60
165/165 ————— 1s 4ms/step - loss: 0.0261 - val_loss: 0.0239
Epoch 5/60
165/165 ————— 1s 4ms/step - loss: 0.0217 - val_loss: 0.0204
Epoch 6/60
165/165 ————— 1s 4ms/step - loss: 0.0184 - val_loss: 0.0172
Epoch 7/60
165/165 ————— 1s 4ms/step - loss: 0.0161 - val_loss: 0.0152
Epoch 8/60
165/165 ————— 1s 4ms/step - loss: 0.0145 - val_loss: 0.0140
Epoch 9/60
165/165 ————— 1s 4ms/step - loss: 0.0135 - val_loss:

```
0.0131
Epoch 10/60
165/165 _____ 1s 4ms/step - loss: 0.0126 - val_loss:
0.0123
Epoch 11/60
165/165 _____ 1s 5ms/step - loss: 0.0119 - val_loss:
0.0120
Epoch 12/60
165/165 _____ 1s 6ms/step - loss: 0.0113 - val_loss:
0.0112
Epoch 13/60
165/165 _____ 1s 6ms/step - loss: 0.0108 - val_loss:
0.0106
Epoch 14/60
165/165 _____ 1s 4ms/step - loss: 0.0104 - val_loss:
0.0103
Epoch 15/60
165/165 _____ 1s 5ms/step - loss: 0.0101 - val_loss:
0.0101
Epoch 16/60
165/165 _____ 1s 4ms/step - loss: 0.0099 - val_loss:
0.0098
Epoch 17/60
165/165 _____ 1s 4ms/step - loss: 0.0096 - val_loss:
0.0098
Epoch 18/60
165/165 _____ 1s 4ms/step - loss: 0.0095 - val_loss:
0.0094
Epoch 19/60
165/165 _____ 1s 4ms/step - loss: 0.0093 - val_loss:
0.0095
Epoch 20/60
165/165 _____ 1s 4ms/step - loss: 0.0092 - val_loss:
0.0092
Epoch 21/60
165/165 _____ 1s 4ms/step - loss: 0.0091 - val_loss:
0.0091
Epoch 22/60
165/165 _____ 1s 3ms/step - loss: 0.0090 - val_loss:
0.0091
Epoch 23/60
165/165 _____ 1s 4ms/step - loss: 0.0090 - val_loss:
0.0091
Epoch 24/60
165/165 _____ 1s 4ms/step - loss: 0.0089 - val_loss:
0.0090
Epoch 25/60
165/165 _____ 1s 4ms/step - loss: 0.0089 - val_loss:
0.0089
```

```
Epoch 26/60
165/165 _____ 1s 4ms/step - loss: 0.0088 - val_loss:
0.0089
Epoch 27/60
165/165 _____ 1s 4ms/step - loss: 0.0088 - val_loss:
0.0090
Epoch 28/60
165/165 _____ 1s 4ms/step - loss: 0.0088 - val_loss:
0.0090
Epoch 29/60
165/165 _____ 1s 6ms/step - loss: 0.0088 - val_loss:
0.0090
Epoch 30/60
165/165 _____ 1s 6ms/step - loss: 0.0088 - val_loss:
0.0091
Epoch 31/60
165/165 _____ 1s 4ms/step - loss: 0.0087 - val_loss:
0.0088
Epoch 32/60
165/165 _____ 1s 4ms/step - loss: 0.0087 - val_loss:
0.0088
Epoch 33/60
165/165 _____ 1s 4ms/step - loss: 0.0087 - val_loss:
0.0087
Epoch 34/60
165/165 _____ 1s 4ms/step - loss: 0.0087 - val_loss:
0.0087
Epoch 35/60
165/165 _____ 1s 4ms/step - loss: 0.0087 - val_loss:
0.0087
Epoch 36/60
165/165 _____ 1s 4ms/step - loss: 0.0086 - val_loss:
0.0087
Epoch 37/60
165/165 _____ 1s 4ms/step - loss: 0.0086 - val_loss:
0.0095
Epoch 38/60
165/165 _____ 1s 3ms/step - loss: 0.0086 - val_loss:
0.0087
Epoch 39/60
165/165 _____ 1s 4ms/step - loss: 0.0086 - val_loss:
0.0086
Epoch 40/60
165/165 _____ 1s 4ms/step - loss: 0.0086 - val_loss:
0.0087
Epoch 41/60
165/165 _____ 1s 4ms/step - loss: 0.0086 - val_loss:
0.0086
Epoch 42/60
```

```
165/165 ————— 1s 4ms/step - loss: 0.0085 - val_loss: 0.0086
Epoch 43/60
165/165 ————— 1s 4ms/step - loss: 0.0085 - val_loss: 0.0086
Epoch 44/60
165/165 ————— 1s 5ms/step - loss: 0.0085 - val_loss: 0.0085
Epoch 45/60
165/165 ————— 1s 5ms/step - loss: 0.0085 - val_loss: 0.0086
Epoch 46/60
165/165 ————— 1s 6ms/step - loss: 0.0085 - val_loss: 0.0086
Epoch 47/60
165/165 ————— 1s 4ms/step - loss: 0.0085 - val_loss: 0.0088
Epoch 48/60
165/165 ————— 1s 4ms/step - loss: 0.0085 - val_loss: 0.0085
Epoch 49/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 50/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 51/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 52/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 53/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 54/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 55/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 56/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0086
Epoch 57/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss: 0.0085
Epoch 58/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss:
```

```
0.0087
Epoch 59/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss:
0.0084
Epoch 60/60
165/165 ————— 1s 4ms/step - loss: 0.0084 - val_loss:
0.0084
Epoch 1/60
165/165 ————— 4s 12ms/step - loss: 0.0962 - val_loss:
0.0530
Epoch 2/60
165/165 ————— 1s 4ms/step - loss: 0.0517 - val_loss:
0.0504
Epoch 3/60
165/165 ————— 1s 4ms/step - loss: 0.0495 - val_loss:
0.0484
Epoch 4/60
165/165 ————— 1s 4ms/step - loss: 0.0474 - val_loss:
0.0455
Epoch 5/60
165/165 ————— 1s 4ms/step - loss: 0.0436 - val_loss:
0.0411
Epoch 6/60
165/165 ————— 1s 4ms/step - loss: 0.0391 - val_loss:
0.0371
Epoch 7/60
165/165 ————— 1s 4ms/step - loss: 0.0359 - val_loss:
0.0348
Epoch 8/60
165/165 ————— 1s 4ms/step - loss: 0.0338 - val_loss:
0.0328
Epoch 9/60
165/165 ————— 1s 4ms/step - loss: 0.0321 - val_loss:
0.0315
Epoch 10/60
165/165 ————— 1s 4ms/step - loss: 0.0311 - val_loss:
0.0309
Epoch 11/60
165/165 ————— 1s 5ms/step - loss: 0.0305 - val_loss:
0.0304
Epoch 12/60
165/165 ————— 1s 7ms/step - loss: 0.0302 - val_loss:
0.0301
Epoch 13/60
165/165 ————— 1s 5ms/step - loss: 0.0300 - val_loss:
0.0300
Epoch 14/60
165/165 ————— 1s 4ms/step - loss: 0.0299 - val_loss:
0.0300
```

```
Epoch 15/60
165/165 _____ 1s 4ms/step - loss: 0.0298 - val_loss:
0.0297
Epoch 16/60
165/165 _____ 1s 4ms/step - loss: 0.0296 - val_loss:
0.0297
Epoch 17/60
165/165 _____ 1s 4ms/step - loss: 0.0294 - val_loss:
0.0293
Epoch 18/60
165/165 _____ 1s 4ms/step - loss: 0.0289 - val_loss:
0.0284
Epoch 19/60
165/165 _____ 1s 4ms/step - loss: 0.0276 - val_loss:
0.0270
Epoch 20/60
165/165 _____ 1s 4ms/step - loss: 0.0265 - val_loss:
0.0263
Epoch 21/60
165/165 _____ 1s 4ms/step - loss: 0.0261 - val_loss:
0.0261
Epoch 22/60
165/165 _____ 1s 4ms/step - loss: 0.0259 - val_loss:
0.0259
Epoch 23/60
165/165 _____ 1s 4ms/step - loss: 0.0258 - val_loss:
0.0258
Epoch 24/60
165/165 _____ 1s 4ms/step - loss: 0.0257 - val_loss:
0.0257
Epoch 25/60
165/165 _____ 1s 4ms/step - loss: 0.0256 - val_loss:
0.0257
Epoch 26/60
165/165 _____ 1s 4ms/step - loss: 0.0256 - val_loss:
0.0256
Epoch 27/60
165/165 _____ 1s 4ms/step - loss: 0.0255 - val_loss:
0.0256
Epoch 28/60
165/165 _____ 1s 6ms/step - loss: 0.0255 - val_loss:
0.0256
Epoch 29/60
165/165 _____ 1s 6ms/step - loss: 0.0255 - val_loss:
0.0256
Epoch 30/60
165/165 _____ 1s 5ms/step - loss: 0.0255 - val_loss:
0.0255
Epoch 31/60
```

```
165/165 ————— 1s 4ms/step - loss: 0.0255 - val_loss: 0.0255
Epoch 32/60
165/165 ————— 1s 4ms/step - loss: 0.0254 - val_loss: 0.0255
Epoch 33/60
165/165 ————— 1s 4ms/step - loss: 0.0254 - val_loss: 0.0255
Epoch 34/60
165/165 ————— 1s 4ms/step - loss: 0.0254 - val_loss: 0.0254
Epoch 35/60
165/165 ————— 1s 4ms/step - loss: 0.0254 - val_loss: 0.0254
Epoch 36/60
165/165 ————— 1s 4ms/step - loss: 0.0253 - val_loss: 0.0253
Epoch 37/60
165/165 ————— 1s 4ms/step - loss: 0.0252 - val_loss: 0.0253
Epoch 38/60
165/165 ————— 1s 4ms/step - loss: 0.0250 - val_loss: 0.0252
Epoch 39/60
165/165 ————— 1s 4ms/step - loss: 0.0248 - val_loss: 0.0246
Epoch 40/60
165/165 ————— 1s 4ms/step - loss: 0.0243 - val_loss: 0.0245
Epoch 41/60
165/165 ————— 1s 4ms/step - loss: 0.0240 - val_loss: 0.0240
Epoch 42/60
165/165 ————— 1s 4ms/step - loss: 0.0239 - val_loss: 0.0241
Epoch 43/60
165/165 ————— 1s 4ms/step - loss: 0.0238 - val_loss: 0.0238
Epoch 44/60
165/165 ————— 1s 7ms/step - loss: 0.0237 - val_loss: 0.0237
Epoch 45/60
165/165 ————— 1s 7ms/step - loss: 0.0236 - val_loss: 0.0237
Epoch 46/60
165/165 ————— 1s 4ms/step - loss: 0.0236 - val_loss: 0.0236
Epoch 47/60
165/165 ————— 1s 4ms/step - loss: 0.0235 - val_loss:
```



```

0.0237
Epoch 48/60
165/165 _____ 1s 4ms/step - loss: 0.0235 - val_loss:
0.0236
Epoch 49/60
165/165 _____ 1s 4ms/step - loss: 0.0235 - val_loss:
0.0235
Epoch 50/60
165/165 _____ 1s 4ms/step - loss: 0.0234 - val_loss:
0.0235
Epoch 51/60
165/165 _____ 1s 4ms/step - loss: 0.0234 - val_loss:
0.0234
Epoch 52/60
165/165 _____ 1s 4ms/step - loss: 0.0234 - val_loss:
0.0234
Epoch 53/60
165/165 _____ 1s 4ms/step - loss: 0.0234 - val_loss:
0.0234
Epoch 54/60
165/165 _____ 1s 4ms/step - loss: 0.0233 - val_loss:
0.0235
Epoch 55/60
165/165 _____ 1s 4ms/step - loss: 0.0233 - val_loss:
0.0234
Epoch 56/60
165/165 _____ 1s 4ms/step - loss: 0.0233 - val_loss:
0.0233
Epoch 57/60
165/165 _____ 1s 4ms/step - loss: 0.0233 - val_loss:
0.0234
Epoch 58/60
165/165 _____ 1s 5ms/step - loss: 0.0233 - val_loss:
0.0234
Epoch 59/60
165/165 _____ 1s 6ms/step - loss: 0.0232 - val_loss:
0.0234
Epoch 60/60
165/165 _____ 1s 7ms/step - loss: 0.0232 - val_loss:
0.0235

```

Model: "deep_convolutional_ae_latent30"

Layer (type) Param #	Output Shape	
image_28x28x1 (InputLayer)	(None, 28, 28, 1)	

0				
		conv2d (Conv2D)	(None, 28, 28, 32)	
320				
		max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	
0				
		conv2d_1 (Conv2D)	(None, 14, 14, 64)	
18,496				
		max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	
0				
		conv2d_2 (Conv2D)	(None, 7, 7, 128)	
73,856				
		flatten (Flatten)	(None, 6272)	
0				
		dense (Dense)	(None, 128)	
802,944				
		latent_30 (Dense)	(None, 30)	
3,870				
		dense_1 (Dense)	(None, 6272)	
194,432				
		reshape (Reshape)	(None, 7, 7, 128)	
0				
		conv2d_transpose	(None, 14, 14, 64)	
73,792		(Conv2DTranspose)		

conv2d_transpose_1	(None, 28, 28, 32)	
18,464		
(Conv2DTranspose)		
linear_decoder (Conv2D)	(None, 28, 28, 1)	
289		

Total params: 1,186,463 (4.53 MB)

Trainable params: 1,186,463 (4.53 MB)

Non-trainable params: 0 (0.00 B)

Epoch 1/50

165/165 ————— 13s 51ms/step - loss: 0.0496 - val_loss: 0.0226

Epoch 2/50

165/165 ————— 3s 19ms/step - loss: 0.0164 - val_loss: 0.0132

Epoch 3/50

165/165 ————— 3s 19ms/step - loss: 0.0121 - val_loss: 0.0115

Epoch 4/50

165/165 ————— 3s 20ms/step - loss: 0.0107 - val_loss: 0.0102

Epoch 5/50

165/165 ————— 3s 20ms/step - loss: 0.0098 - val_loss: 0.0097

Epoch 6/50

165/165 ————— 3s 19ms/step - loss: 0.0091 - val_loss: 0.0097

Epoch 7/50

165/165 ————— 3s 19ms/step - loss: 0.0087 - val_loss: 0.0085

Epoch 8/50

165/165 ————— 3s 20ms/step - loss: 0.0084 - val_loss: 0.0083

Epoch 9/50

165/165 ————— 3s 20ms/step - loss: 0.0082 - val_loss: 0.0082

Epoch 10/50

165/165 ————— 3s 19ms/step - loss: 0.0081 - val_loss: 0.0081

Epoch 11/50

165/165 ————— 3s 19ms/step - loss: 0.0079 - val_loss: 0.0087

```
Epoch 12/50
165/165 ————— 3s 20ms/step - loss: 0.0078 - val_loss: 0.0078
Epoch 13/50
165/165 ————— 3s 19ms/step - loss: 0.0078 - val_loss: 0.0078
Epoch 14/50
165/165 ————— 3s 19ms/step - loss: 0.0077 - val_loss: 0.0078
Epoch 15/50
165/165 ————— 3s 19ms/step - loss: 0.0077 - val_loss: 0.0078
Epoch 16/50
165/165 ————— 4s 21ms/step - loss: 0.0077 - val_loss: 0.0077
Epoch 17/50
165/165 ————— 3s 19ms/step - loss: 0.0076 - val_loss: 0.0078
Epoch 18/50
165/165 ————— 5s 19ms/step - loss: 0.0076 - val_loss: 0.0077
Epoch 19/50
165/165 ————— 3s 20ms/step - loss: 0.0076 - val_loss: 0.0080
Epoch 20/50
165/165 ————— 5s 19ms/step - loss: 0.0075 - val_loss: 0.0077
Epoch 21/50
165/165 ————— 3s 19ms/step - loss: 0.0075 - val_loss: 0.0077
Epoch 22/50
165/165 ————— 3s 19ms/step - loss: 0.0074 - val_loss: 0.0077
Epoch 23/50
165/165 ————— 3s 20ms/step - loss: 0.0074 - val_loss: 0.0077
Epoch 24/50
165/165 ————— 3s 19ms/step - loss: 0.0074 - val_loss: 0.0076
Epoch 25/50
165/165 ————— 3s 19ms/step - loss: 0.0074 - val_loss: 0.0078
Epoch 26/50
165/165 ————— 3s 19ms/step - loss: 0.0074 - val_loss: 0.0076
Epoch 27/50
165/165 ————— 3s 20ms/step - loss: 0.0073 - val_loss: 0.0079
Epoch 28/50
```

```
165/165 ————— 3s 19ms/step - loss: 0.0073 - val_loss: 0.0076
Epoch 29/50
165/165 ————— 3s 19ms/step - loss: 0.0073 - val_loss: 0.0076
Epoch 30/50
165/165 ————— 3s 20ms/step - loss: 0.0072 - val_loss: 0.0076
Epoch 31/50
165/165 ————— 3s 19ms/step - loss: 0.0073 - val_loss: 0.0076
Epoch 32/50
165/165 ————— 3s 19ms/step - loss: 0.0072 - val_loss: 0.0075
Epoch 33/50
165/165 ————— 3s 19ms/step - loss: 0.0073 - val_loss: 0.0076
Epoch 34/50
165/165 ————— 3s 20ms/step - loss: 0.0072 - val_loss: 0.0077
Epoch 35/50
165/165 ————— 3s 19ms/step - loss: 0.0072 - val_loss: 0.0075
Epoch 36/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss: 0.0075
Epoch 37/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss: 0.0081
Epoch 38/50
165/165 ————— 4s 21ms/step - loss: 0.0072 - val_loss: 0.0075
Epoch 39/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss: 0.0077
Epoch 40/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss: 0.0076
Epoch 41/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss: 0.0075
Epoch 42/50
165/165 ————— 3s 20ms/step - loss: 0.0071 - val_loss: 0.0075
Epoch 43/50
165/165 ————— 3s 19ms/step - loss: 0.0070 - val_loss: 0.0075
Epoch 44/50
165/165 ————— 5s 19ms/step - loss: 0.0070 - val_loss:
```

```

0.0076
Epoch 45/50
165/165 ————— 3s 19ms/step - loss: 0.0071 - val_loss:
0.0075
Epoch 46/50
165/165 ————— 3s 19ms/step - loss: 0.0070 - val_loss:
0.0081
Epoch 47/50
165/165 ————— 3s 19ms/step - loss: 0.0070 - val_loss:
0.0075
Epoch 48/50
165/165 ————— 3s 19ms/step - loss: 0.0070 - val_loss:
0.0076
Epoch 49/50
165/165 ————— 3s 19ms/step - loss: 0.0070 - val_loss:
0.0076
Epoch 50/50
165/165 ————— 3s 19ms/step - loss: 0.0069 - val_loss:
0.0075

```

```

{"summary":{"\n  \"name\": \"task3_results\",\n  \"rows\": 6,\n  \"fields\": [\n    {\n      \"column\": \"dataset\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"CIFAR10-gray\",\n          \"MNIST\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"method\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 3,\n        \"samples\": [\n          \"single hidden dense AE, sigmoid encoder\n30, linear decoder\",\n          \"three hidden dense AE, sigmoid 10-10-10, linear decoder\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"average_test_snr_db\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 4.6041088157563825,\n        \"min\": 9.667779713860803,\n        \"max\": 20.967988979958356,\n        \"num_unique_values\": 6,\n        \"samples\": [\n          12.95055592444497,\n          9.667779713860803\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}, \"type\": \"dataframe\", \"variable_name\": \"task3_results\"}

```

Direct answer: average SNR when 30 nodes split equally as 10-10-10 across 3 hidden layers

```

print('Task 3 answer: average test SNR with 10-10-10 hidden layer
distribution (sigmoid hidden, linear decoder):')
print()
for _, row in
task3_results[task3_results['method'].str.contains('three
hidden')].iterrows():
    print(f' {row["dataset"]}: {row["average_test_snr_db"]:.2f} dB')

```

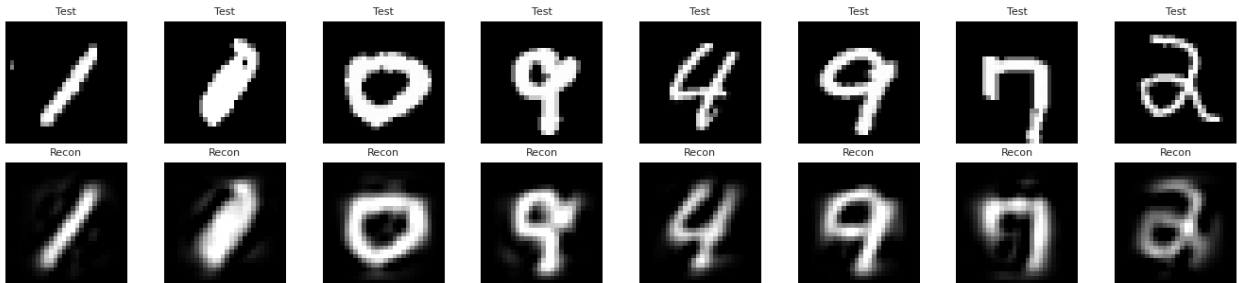
Task 3 answer: average test SNR with 10-10-10 hidden layer distribution (sigmoid hidden, linear decoder):

MNIST: 9.67 dB

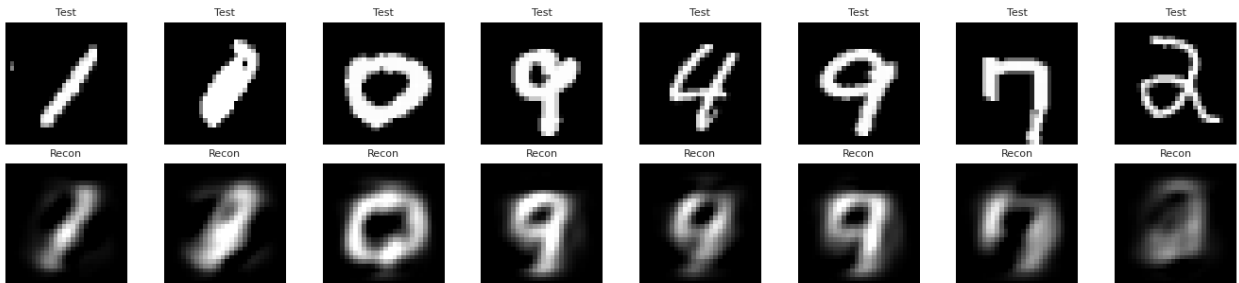
CIFAR10-gray: 15.50 dB

```
for ds in datasets:
    for key, title in [
        ("single", "single hidden dense AE"),
        ("three", "three hidden dense AE, 10-10-10"),
        ("conv", "deep convolutional AE, latent=30"),
    ]:
        recon = task3_models[(ds.name, key)]["recon_50_200"]
        plot_reconstruction_examples(ds.x_test_flat_50_200, recon,
f"{ds.name}: {title} reconstructions")
```

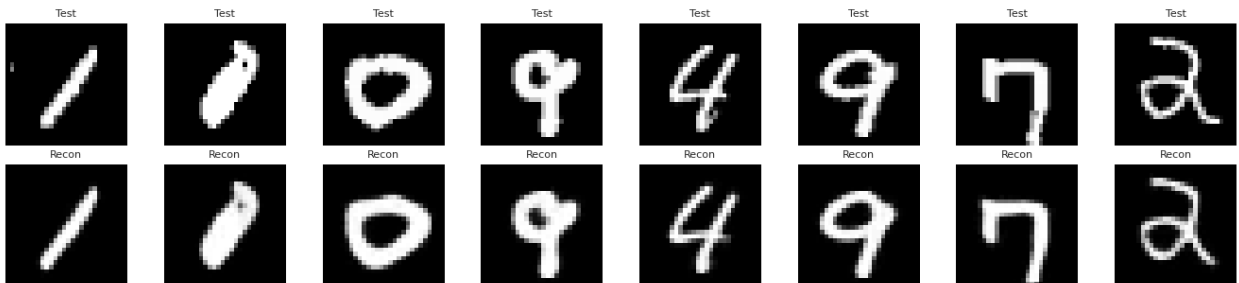
MNIST: single hidden dense AE reconstructions



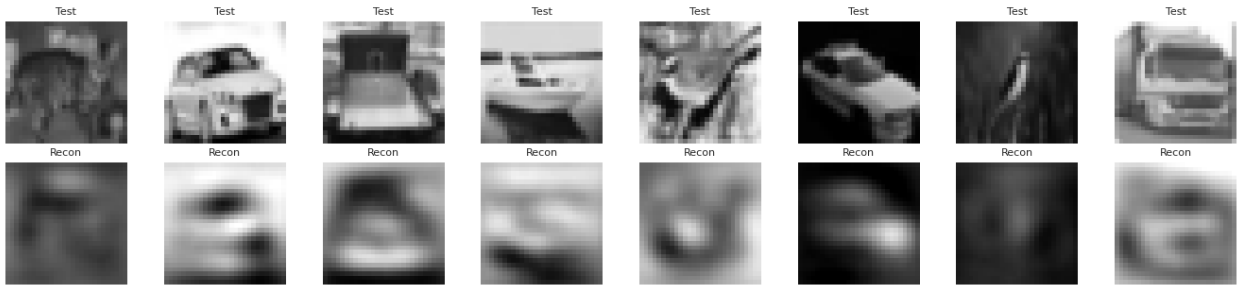
MNIST: three hidden dense AE, 10-10-10 reconstructions



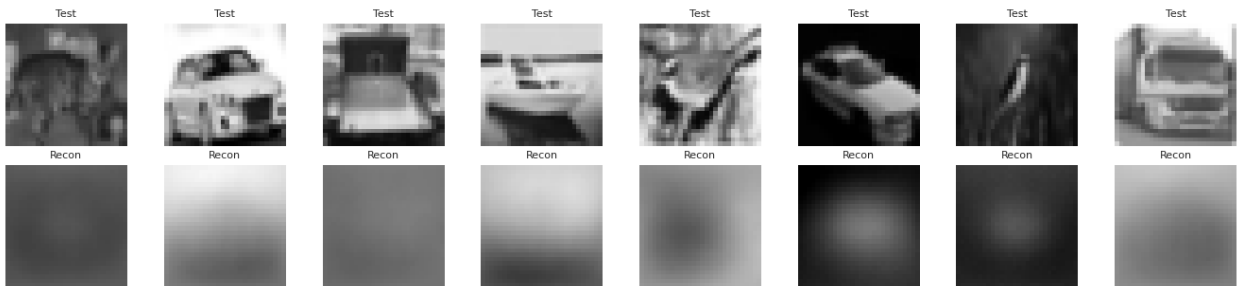
MNIST: deep convolutional AE, latent=30 reconstructions



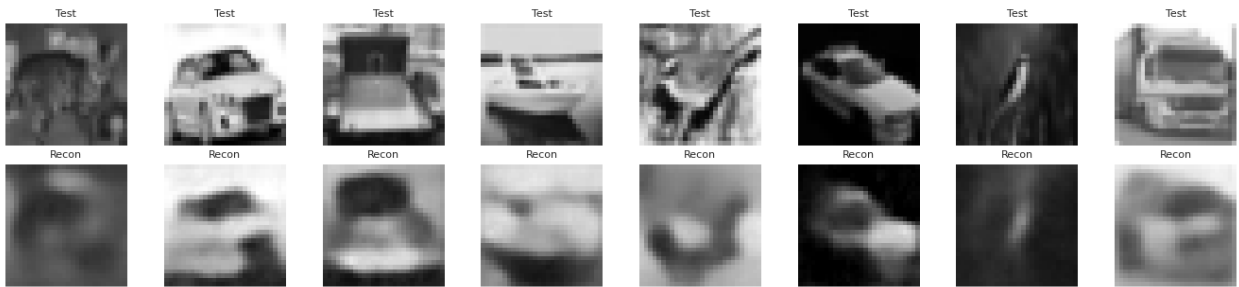
CIFAR10-gray: single hidden dense AE reconstructions



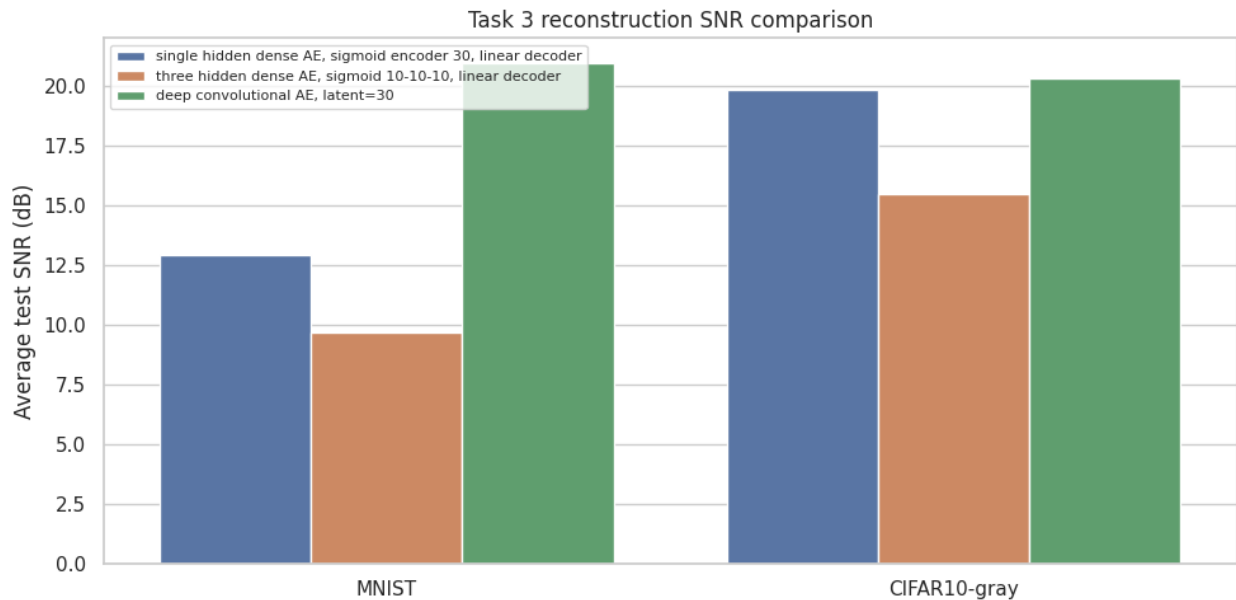
CIFAR10-gray: three hidden dense AE, 10-10-10 reconstructions



CIFAR10-gray: deep convolutional AE, latent=30 reconstructions



```
plt.figure(figsize=(10, 5))
sns.barplot(data=task3_results, x="dataset", y="average_test_snr_db",
            hue="method")
plt.title("Task 3 reconstruction SNR comparison")
plt.ylabel("Average test SNR (dB)")
plt.xlabel("")
plt.xticks(rotation=0)
plt.legend(loc="best", fontsize=8)
plt.tight_layout()
plt.show()
```

Task 3 comments.

The convolutional autoencoder is expected to achieve the highest SNR because convolutional filters exploit spatial locality and weight sharing. The single hidden layer dense autoencoder with only 30 sigmoid hidden nodes is a much stronger bottleneck and ignores image geometry. The three-hidden-layer dense model uses the same total hidden-node budget split as 10-10-10, so the effective bottleneck is only 10 dimensions; it normally reconstructs worse than the 30-node single hidden layer model, especially for gray CIFAR10. MNIST is usually easier to reconstruct than CIFAR10 because digits are sparse and structurally simpler.

Final Summary Tables

```
print("Task 1: PCA and randomized PCA")
display(task1_results)

print("Task 1: top 30 eigenvalues")
display(task1_top30_eigenvalues)

print("Task 2: PCA versus tied linear AE classification and SNR")
display(
    pd.concat(
        [
            task1_results.query("method == 'standard PCA'")[
                ["dataset", "method", "components", "accuracy",
                "macro_auc_ovr", "average_test_snr_db"]
            ],
            task2_results,
        ],
        ignore_index=True,
    )
)
```

```
)
```

```
print("Task 2: quantitative component relationship")
display(component_comparison)
```

```
print("Task 3: nonlinear autoencoder SNR")
display(task3_results)
```

Task 1: PCA and randomized PCA

```
{"summary":{"\n  \"name\": \"task1_results\",\n  \"rows\": 4,\n  \"fields\": [\n    {\n      \"column\": \"dataset\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"CIFAR10-gray\",\n          \"MNIST\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"method\",\n        \"properties\": {\n          \"dtype\": \"string\",\n          \"num_unique_values\": 2,\n          \"samples\": [\n            \"randomized PCA\",\n            \"standard PCA\"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"components\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 0,\n            \"min\": 30,\n            \"max\": 30,\n            \"num_unique_values\": 1,\n            \"samples\": [\n              30\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\": \"accuracy\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 0.33856783221752135,\n              \"min\": 0.307,\n              \"max\": 0.8935714285714286,\n              \"num_unique_values\": 4,\n              \"samples\": [\n                0.8934285714285715\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n            \"column\": \"macro_auc_ovr\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 0.15039327267694105,\n              \"min\": 0.7296972530864198,\n              \"max\": 0.9902317998587511,\n              \"num_unique_values\": 4,\n              \"samples\": [\n                0.9901408556435897\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n            \"column\": \"explained_variance_ratio_sum\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 0.06820361289378955,\n              \"min\": 0.7314825654029846,\n              \"max\": 0.8496577143669128,\n              \"num_unique_values\": 4,\n              \"samples\": [\n                0.7314825654029846\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n            \"column\": \"average_test_snr_db\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 4.289376589612675,\n              \"min\": 12.571396940483252,\n              \"max\": 20.00122318203906,\n              \"num_unique_values\": 4,\n              \"samples\": [\n                12.571396940483252\n              ],\n              \"semantic_type\": \"\",
```

```
\n"description\\": \\\"\\\"\\n      }\\n    }\\n  ]\\n}\\n\", \"type\": \"dataframe\", \"variable_name\": \"task1_results\"}
```

Task 1: top 30 eigenvalues

```
{\"summary\": \"{\\n  \\\"name\\\": \\\"task1_top30_eigenvalues\\\",\\n  \\\"rows\\\": 120,\\n  \\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"dataset\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"category\\\",\\n        \\\"num_unique_values\\\": 2,\\n        \\\"samples\\\": [\\n          \\\"CIFAR10-gray\\\",\\n          \\\"MNIST\\\"\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"method\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"category\\\",\\n        \\\"num_unique_values\\\": 2,\\n        \\\"samples\\\": [\\n          \\\"randomized PCA\\\",\\n          \\\"standard PCA\\\"\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"principal_component\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"number\\\",\\n        \\\"std\\\": 8,\\n        \\\"min\\\": 1,\\n        \\\"max\\\": 30,\\n        \\\"num_unique_values\\\": 30,\\n        \\\"samples\\\": [\\n          28,\\n          16\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"eigenvalue\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"float32\\\",\\n        \\\"num_unique_values\\\": 118,\\n        \\\"samples\\\": [\\n          9337.3583984375,\\n          301781.0\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"explained_variance_ratio\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"float32\\\",\\n        \\\"num_unique_values\\\": 119,\\n        \\\"samples\\\": [\\n          0.008097657933831215,\\n          0.0034726159647107124\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    }\\n  ]\\n}\\n\", \"type\": \"dataframe\", \"variable_name\": \"task1_top30_eigenvalues\"}
```

Task 2: PCA versus tied linear AE classification and SNR

```
{\"summary\": \"{\\n  \\\"name\\\": \\\"display(task3_results)\\\",\\n  \\\"rows\\\": 4,\\n  \\\"fields\\\": [\\n    {\\n      \\\"column\\\": \\\"dataset\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"string\\\",\\n        \\\"num_unique_values\\\": 2,\\n        \\\"samples\\\": [\\n          \\\"CIFAR10-gray\\\",\\n          \\\"MNIST\\\"\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"method\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"string\\\",\\n        \\\"num_unique_values\\\": 2,\\n        \\\"samples\\\": [\\n          \\\"tied linear AE features\\\",\\n          \\\"standard PCA\\\"\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\": \\\"components\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"number\\\",\\n        \\\"std\\\": 0,\\n        \\\"min\\\": 30,\\n        \\\"max\\\": 30,\\n        \\\"num_unique_values\\\": 1,\\n        \\\"samples\\\": [\\n          30\\n        ],\\n        \\\"semantic_type\\\": \\\"\\\",\\n        \\\"description\\\": \\\"\\\"\\n      }\\n    }\\n  ]\\n}\\n\", \"type\": \"dataframe\", \"variable_name\": \"task2_results\"}
```

```

"description\": \"\n      }\n    },\n    {\n      \"column\":
\"accuracy\", \n      \"properties\": {\n        \"dtype\":
\"number\", \n        \"std\": 0.33835006011566204, \n        \"min\":
0.3055, \n        \"max\": 0.8935714285714286, \n        \"num_unique_values\": 4, \n        \"samples\": [\n          0.307\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\n      }\n    }, \n    {\n      \"column\": \"macro_auc_ovr\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
0.15041178843338182, \n        \"min\": 0.7293561419753085, \n        \"max\": 0.9902317998587511, \n        \"num_unique_values\": 4, \n        \"samples\": [\n          0.7296978395061728\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\n      }\n    }, \n    {\n      \"column\": \"average_test_snr_db\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\":
4.708006104977068, \n        \"min\": 11.217111564610349, \n        \"max\": 20.00122318203906, \n        \"num_unique_values\": 4, \n        \"samples\": [\n          20.00122318203906\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\n      }\n    }\n  ],\n  \"type\": \"dataframe\"}

```

Task 2: quantitative component relationship

```

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```

```

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```

Task 3: nonlinear autoencoder SNR

```

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```

```

def print_final_observations():
    print("Final observations generated from the current run")
    for ds_name in task1_results["dataset"].unique():
        std = task1_results[(task1_results["dataset"] == ds_name) &
(task1_results["method"] == "standard PCA")].iloc[0]
        rnd = task1_results[(task1_results["dataset"] == ds_name) &
(task1_results["method"] == "randomized PCA")].iloc[0]
        ae = task2_results[task2_results["dataset"] ==
ds_name].iloc[0]
        comp = component_comparison[component_comparison["dataset"] ==
ds_name].iloc[0]
        single = task3_results[(task3_results["dataset"] == ds_name) &
task3_results["method"].str.contains("single hidden",
regex=False)].iloc[0]
        three = task3_results[(task3_results["dataset"] == ds_name) &

```

```

task3_results["method"].str.contains("three hidden",
regex=False)].iloc[0]
    conv = task3_results[(task3_results["dataset"] == ds_name) &
task3_results["method"].str.contains("convolutional",
regex=False)].iloc[0]

    print(f"\n{ds_name}")
    print(f"    Task 1 PCA          : acc={std['accuracy']:.4f}, macro
AUC={std['macro_auc_ovr']:.4f}, SNR={std['average_test_snr_db']:.2f}
dB")
    print(f"    Task 1 randomized: acc={rnd['accuracy']:.4f}, macro
AUC={rnd['macro_auc_ovr']:.4f}, SNR={rnd['average_test_snr_db']:.2f}
dB")
    print(f"    Task 2 tied AE    : acc={ae['accuracy']:.4f}, macro
AUC={ae['macro_auc_ovr']:.4f}, SNR={ae['average_test_snr_db']:.2f}
dB")
    print(f"    PCA-AE relation  : mean matched |cos|
={comp['mean_matched_abs_cosine']:.4f}, max principal
angle={comp['max_principal_angle_deg']:.2f} degrees")
    print(f"    Task 3 single AE :
SNR={single['average_test_snr_db']:.2f} dB")
    print(f"    Task 3 10-10-10 :
SNR={three['average_test_snr_db']:.2f} dB")
    print(f"    Task 3 conv AE  :
SNR={conv['average_test_snr_db']:.2f} dB")

    print("\nConclusion: randomized PCA should closely match standard
PCA; the tied linear AE should recover a PCA-like subspace; "
        "and the convolutional AE is expected to reconstruct best
because it uses spatial structure.")

```

```
print_final_observations()
```

Final observations generated from the current run

MNIST

```

Task 1 PCA          : acc=0.8936, macro AUC=0.9902, SNR=12.57 dB
Task 1 randomized: acc=0.8934, macro AUC=0.9901, SNR=12.57 dB
Task 2 tied AE     : acc=0.8910, macro AUC=0.9899, SNR=11.22 dB
PCA-AE relation    : mean matched |cos|=0.0059, max principal
angle=89.98 degrees
Task 3 single AE   : SNR=12.95 dB
Task 3 10-10-10    : SNR=9.67 dB
Task 3 conv AE     : SNR=20.97 dB

```

CIFAR10-gray

```

Task 1 PCA          : acc=0.3070, macro AUC=0.7297, SNR=20.00 dB
Task 1 randomized: acc=0.3072, macro AUC=0.7297, SNR=20.00 dB
Task 2 tied AE     : acc=0.3055, macro AUC=0.7294, SNR=19.98 dB

```

PCA-AE relation : mean matched $|\cos|=0.3954$, max principal
angle=32.39 degrees

Task 3 single AE : SNR=19.84 dB

Task 3 10-10-10 : SNR=15.50 dB

Task 3 conv AE : SNR=20.32 dB

Conclusion: randomized PCA should closely match standard PCA; the tied linear AE should recover a PCA-like subspace; and the convolutional AE is expected to reconstruct best because it uses spatial structure.